

Optimal Transit Networks in Developing Countries*

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May 2026

Abstract

A key allocation problem for budget-constrained governments in developing cities is where and how much public mass transit infrastructure to build, relative to allowing privatized and often informal transit operators to operate. Mass transit technologies are fixed-cost intensive but have low marginal operating costs, while minibuses have higher marginal costs and entail externalities, but require little upfront investment. In this paper we study the optimal transit network problem in general spatial equilibrium, where a planner can choose between technologies to connect locations within a city. We show that the optimal public-private transit mix on any given edge of the network depends crucially on the trade-off between relative marginal and fixed costs across technologies. Leveraging a unique dataset of public and private transit networks in Mexico City and microdata on wages, land use, and commuting flows, we quantitatively study the gains from budget-feasible expansions of the transit system. We find that increasing the infrastructure budget by 50% raises welfare by 1.33% under public-only expansion and by 1.59% when both public and private transit can expand optimally, with the additional 20% welfare gain driven by private transit. In both cases, optimal investments primarily improve connections from peripheral areas to productive outer nodes and toward the existing network. These results suggest that developing cities could benefit substantially by jointly designing public and private networks, rather than public mass transit in isolation.

*We are deeply grateful to Santiago Veleza for outstanding research assistance. We also acknowledge and appreciate help from Claude Code and Codex.

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1 INTRODUCTION

Urban transit systems in developing cities face a fundamental allocation problem: how scarce public resources should be allocated between publicly provided mass transit and privatized, often informal, transit provision. This question is quantitatively important: privately operated minibuses today provide between 50 and 100 percent of urban transit in many developing-world cities ([Conwell, 2024](#)), yet the optimal design of mixed public-private transit networks remains poorly understood. Different allocation choices can have profound implications for urban structure, welfare, and commuting patterns by shaping congestion, environmental externalities, and the overall cost of commuting between any two locations ([Mosqueda, 2026](#)). These issues are particularly salient in developing countries, where rapid urbanization and tight fiscal constraints limit governments’ ability to expand mass transit networks.

The central question of this paper is how a planner should allocate public and private investment in a network. The planner faces a key trade-off. Mass transit technologies are fixed-cost intensive, requiring large upfront investments but featuring low marginal operating costs. By contrast, private transit has higher marginal operating costs and entails externalities such as congestion, but requires little upfront investment. When governments cannot afford to extend mass transit to the entire city, these second-best technologies become an attractive alternative to connect underserved areas and meet mobility needs.

Despite its policy relevance, addressing this question poses three main challenges. First, the transit network design problem is computationally intractable: the solution space grows exponentially with city size, and the problem exhibits both substitution and complementarity forces—as routes may complement each other when they connect, but substitute when they compete for riders—which can preclude finding a global optimum ([Kreindler et al., 2023](#)). Second, observed transit networks emerge from decentralized decisions by commuters and private operators, so evaluating optimal policy requires a framework that can jointly account for network design, commuting behavior, and general equilibrium responses. Third, data on actual private routes are scarce throughout the developing world. In the majority of cities, service is informal; therefore, trip geographies and service characteristics are rarely collected centrally.

In this paper we address these challenges by, first, providing a general spatial equilibrium framework that delivers the planner’s optimal public and private networks, and, second, quantifying it by leveraging a unique dataset that describes the universe of public and private lines in the Metropolitan Area of Mexico City. We then use this quantitative model to study where and how much a planner should invest in these two technologies. Counterfactual analyses show that optimally expanding the network using both public and private transit can deliver additional welfare gains of around 18% on top of the gains generated by expanding the public network in isolation, suggesting that developing cities could benefit substantially from joint planning of public and private networks.

Framework. We extend the optimal transport network framework of [Fajgelbaum and Schaal \(2020\)](#), which studies goods trade across regions, to within-city commuting with heterogeneous transit technologies. In our model, the planner chooses i) optimal consumption and labor allocations across space, ii) commuting flows on each edge, and iii) the mix of public and private infrastructure on each link, subject to land, production, and infrastructure budget constraints. The main innovation is to allow each edge to be served by different transport technologies that differ in both operating and construction costs. We assume that commuting costs are a function of technology-specific shifters that capture differences in marginal costs of service provision, which reflects the fact that transporting a worker by subway is substantially cheaper than by minibuses. We also assume that the planner faces a network-building constraint in which she allocates a fixed budget across technologies with different construction costs—reflecting that drilling a tunnel is more expensive than adapting street curbs.

We understand private infrastructure as a bundle that includes the minibuses themselves, service capacity, operating permits, curbs, stops, terminals, and dedicated or priority lane access, while public infrastructure refers to fixed mass-transit assets such as subway, BRT, and rail capacity.

The key modeling choice is that transport costs depend on a continuous mix of technologies, which preserves convexity and makes the problem tractable. Under a log-linear specification of commuting costs, we show that the optimal public-private infrastructure mix on each edge depends exclusively on the trade-off between the relative marginal cost of operation and construction across technologies: even if public transit is more efficient to operate, edges where tunneling is costly, for example, will optimally rely on private provision.

Quantification. The Metropolitan Area of Mexico City provides a natural laboratory to study these trade-offs. Its transit system combines extensive formal public infrastructure with a large private sector, operates under binding fiscal constraints, and exhibits strong complementarities between public and private provision—three features that are central to the model. Private transit is quantitatively dominant: more than half of all trips are made using a minibus.¹ This extensive private network plays a central role in connecting peripheral areas largely underserved by mass transit infrastructure, reflecting the high fixed costs and fiscal constraints that limit public system expansion.

Public and private transit are also deeply intertwined along commuting paths: the majority of transit users rely on private transit for at least one leg of their journey, and most of these trips combine private and public services ([Mosqueda, 2026](#)). At the same time, private provision generates congestion externalities, with more than 80 private lines overlapping on many road links ([Mosqueda, 2026](#)). Together, these features make Mexico City a particularly informative setting to study how a planner should optimally allocate public and private transit technologies across a network, and how their interaction shapes commuting costs, congestion, and welfare in equilibrium.

¹We calculate this from a travel survey, INEGI Encuesta Origen-Destino 2017.

To bring the model to the data, we combine a unique set of high-resolution data sources for Mexico City that jointly characterize the transit network, commuting environment, and spatial economic structure. The core input is a detailed General Transit Feed Specification (GTFS) collected by the private firm “Where Is My Transport”. This dataset provides a unified representation of the city’s full transit system, including both formal public mass transit—metro, BRT, and suburban rail—and private informal services such as *combis* and *micros*, with complete route-level information on network topology, frequencies, and travel times.

To motivate the analysis, we use these data to document two sets of facts about Mexico City’s transit system. Private transit accounts for 85 percent of all route-kilometers in the network—nearly six times the coverage of public mass transit—and this share rises from 68 percent near the CBD to 100 percent in the outermost periphery. Public and private modes are not substitutes but are tightly linked within individual trips: 33 percent of all transit commutes combine both modes in a single journey, and combi-to-metro transitions alone make up half of all mode-to-mode switches. On average, commuters on these mixed trips spend 37 minutes on private transit for access and egress—comparable to the in-vehicle time on public segments—and 81 percent of metro stations have at least one combi stop within 100 meters. These patterns show that private transit serves a structural first-and-last-mile role, so that any evaluation of public investment must account for how it reshapes the private trips that feed into it.

We then calibrate our model. We use the GTFS dataset to construct a simplified transit network that consists of 427 nodes and 1,187 edges. Each node in the network is richly characterized. We leverage spatially granular microdata on wages, population, employment, and commuting flows from INEGI, as well as detailed land-use information from the Secretaría de Desarrollo Urbano. Together, the network and administrative microdata allow us to calibrate the two key primitives of the model: mode-specific operating costs and infrastructure construction costs. We calibrate the building and operating cost parameters to the equilibrium in which the population distribution and the network are fixed to the observed ones. We identify higher fixed investment costs to build public transit relative to private transit, but lower marginal operating costs. The calibrated framework thus enables us to quantitatively evaluate counterfactual infrastructure allocations.

Welfare effects of transit expansions. We study a set of counterfactual exercises to assess the gains from budget-feasible expansions of the transit system. The counterfactuals differ in the extent to which the government accounts for private transport infrastructure when designing the network. In particular, we consider two cases: i) one where the planner expands the network using only public transit, and ii) another where the planner considers both public and private investments. In both scenarios, we consider a 50% infrastructure budget increase and compare the two scenarios against the benchmark observed equilibrium where the network is fixed to the observed one.

We show that welfare gains from network expansion depend critically on how private transit is incorporated into the planning problem. When public infrastructure is designed without accounting for joint optimal expansions of the private network, the resulting public-only ex-

pansions deliver smaller welfare gains. We find that optimally expanding the public network alone delivers a 1.33% welfare gain, but the welfare gain under joint public-private planning is approximately 20% larger, reaching 1.59%. These results highlight that a significant share of the potential benefits from transit investment arises not from expanding public infrastructure alone, but from integrating private transport into network design decisions.

In terms of the spatial distribution of investments, we find that both public and private investments aim to better connect previously poorly connected productive hubs located on the outskirts of the city. We also show that new investments attempt to connect peripheral nodes to one another and to the existing transit infrastructure, which is predominantly concentrated in central areas of the city.

Taken together, the paper provides a unified framework for analyzing and computing optimal transit networks in budget-constrained cities. Rather than treating informal transit as a temporary fix or a policy failure, the analysis shows how its provision emerges endogenously from the planner’s optimal choices and how its interaction with public transit shapes urban structure, commuting flows, and welfare.

Related literature. This paper relates to a growing literature on optimal transportation networks and infrastructure investment in general equilibrium. Several contributions study network design by optimizing over transport links or transport costs ([Alder, 2025](#); [Tarasov and Felbermayr, 2014](#); [Allen and Arkolakis, 2022](#)), typically treating transport technologies or bilateral costs as primitives and focusing on marginal or decentralized adjustments around an existing network. Our problem is also conceptually related to optimal transport network design in non-economic settings, which studies network formation under flow and cost constraints ([Bernot et al., 2009](#)). More closely related are recent papers that analyze optimal transport infrastructure and policy in spatial equilibrium models with endogenous flows and congestion ([Felbermayr and Tarasov, 2022](#); [Allen and Arkolakis, 2022](#); [Almagro et al., 2024](#)). Closest to our approach, [Fajgelbaum and Schaal \(2020\)](#) develops a quantitative framework for optimal transport networks in spatial equilibrium that allows for network-wide welfare evaluation; we build on their framework and extend it to a setting with heterogeneous transport technologies, adapting it from goods trade to within-city commuting and to the coexistence of public mass transit and decentralized private transit.

This paper relates to the literature on urban transit infrastructure and welfare in developing cities. More broadly, our analysis connects to the literature on aggregate welfare losses from misallocation ([Rogerson and Restuccia, 2004](#); [Hsieh and Klenow, 2009](#)) and to work emphasizing spatial misallocation arising from geographic frictions or spatially targeted policies ([Desmet and Rossi-Hansberg, 2013](#); [Adamopoulos et al., 2022](#); [Asturias et al., 2014](#); [Fajgelbaum et al., 2019](#); [Hsieh and Moretti, 2019](#)). Within the urban context, existing work provides causal evidence that large-scale public transit investments can generate substantial welfare gains in congested urban environments, primarily through reductions in commuting costs and changes in residential sorting ([Tsivanidis, 2019](#)). Complementary studies evaluate the welfare effects of specific single-mode, publicly provided transit expansions, including cable car investments

in Medellín (Khanna et al., 2024) and subway expansions in Mexico City (Zárate, 2024). In contrast to this literature, which typically evaluates isolated public transit projects or abstracts from the coexistence of multiple transport modes, our contribution is to study welfare and misallocation in a unified framework where public mass transit and decentralized private or informal transit jointly determine commuting patterns and welfare under binding fiscal constraints.

A large body of research studies how changes in transportation infrastructure and transport costs affect economic activity and productivity. Some papers estimate the effects of road expansion on productivity and industrial outcomes in the United States (Fernald, 1999), regional economic activity along highway corridors (Chandra and Thompson, 2000; Baum-Snow, 2007; Duranton et al., 2014), access to railways in India (Donaldson, 2018), and expressway construction in China (Faber, 2014). This literature highlights the central role of transportation networks in shaping regional growth and spatial economic outcomes. Our approach complements these studies by focusing on the optimal allocation of transit technologies within cities rather than on realized expansions of a single mode, allowing us to quantify how infrastructure choices interact with population distribution, congestion, and economic activity in general equilibrium.

This paper also contributes to the literature on the coexistence of formal and informal urban transport systems in developing cities. Recent work shows that public transit investments interact strongly with incumbent private transit, shaping prices, waiting times, entry, and welfare through equilibrium responses of private operators (Conwell, 2024; Björkegren et al., 2025). In particular, public entry can crowd out or reallocate private supply, with welfare effects that depend on commuters’ tradeoffs between prices and waiting times and on private operators’ responses along connected routes (Björkegren et al., 2025), while frictions internal to informal transit—such as queuing, scale economies, and market power—play a central role in determining equilibrium outcomes and policy effectiveness (Conwell, 2024). Closely related work examines optimal public transport networks and commuters’ responses to waiting times and service frequency in large developing-country cities, abstracting from informal provision while highlighting the importance of system-wide equilibrium forces (Kreindler et al., 2023). We complement this literature by developing a quantitative spatial equilibrium model in which a budget-constrained planner chooses the optimal allocation of public mass transit and decentralized private/informal transit; the planner’s solution is disciplined by an estimated elasticity of substitution between modes that governs their complementarity in equilibrium.

2 MODEL

We build on Fajgelbaum and Schaal (2020) and extend their framework for optimal road networks and trade between locations to a city setting with commuting and transit networks. The key difference is that the planner chooses mode-specific flows and mode-specific infrastructure, particularly public and private transit infrastructure, so an edge in the network is no

longer summarized by one aggregate commuting flow and one infrastructure measure. The output of the planner's problem thus gives an allocation of residents, workers, and mode-specific flows and infrastructure across locations.

Environment. The economy consists of a discrete set of locations $\mathcal{J} = \{1, \dots, J\}$ and has a fixed number of workers of measure $\bar{L}$. Locations are arranged on a directed graph $(\mathcal{J}, \mathcal{E})$, where $\mathcal{E}$ denotes the set of edges. Workers may live and work in different locations because they can commute. The mass of workers living and working in a given location j is denoted by L_j^R and L_j^L , respectively. Worker commuting between locations generates a commuting flow L_{jk} across any ordered node pair (j, k) . Workers can only travel through connected locations. A connection is defined by the neighbor set of nodes; that is, any location j has neighbors $k \in \mathcal{N}(j)$. To reach any $k' \notin \mathcal{N}(j)$, workers must travel through a sequence of connected locations.

Let $\mathcal{M} \equiv \{\text{pub}, \text{pri}\}$ denote the set of transit technologies. For each edge $(j, k) \in \mathcal{E}$ and mode $m \in \mathcal{M}$, let $L_{jk}^m \geq 0$ denote the commuting flow from j to k using mode m . The aggregate flow on the physical edge is

$$L_{jk} \equiv L_{jk}^{\text{pub}} + L_{jk}^{\text{pri}}. \quad (1)$$

Analogously, the aggregate infrastructure on each edge is composed of public and private infrastructure:

$$I_{jk} \equiv I_{jk}^{\text{pub}} + I_{jk}^{\text{pri}}. \quad (2)$$

The planner therefore chooses not only how many workers move across each edge, but also which technology or type of investment to use on that edge.

Transport technology and commuting costs. Commuting consumes real resources, measured in units of labor. One can think of this as a literal loss of the daily time endowment, exhaustion, fatigue, or any other aspect of commuting that results in a loss of productive units of labor. Let T_{jk}^m denote the labor-equivalent commuting resource cost generated by mode- m flow on edge (j, k) .

We assume that private transit is congestible. This captures the idea that, in practice, lower private-transit capacity implies longer waiting times because of boarding and matching externalities (Conwell, 2024). Let I_{jk}^{pri} denote private-transit infrastructure on edge (j, k) . Following Fajgelbaum and Schaal (2020), we parameterize the private commuting resource cost on link (j, k) as

$$T_{jk}^{\text{pri}} = \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}. \quad (3)$$

Here $\delta_{jk}^{\tau, \text{pri}}$ captures geographic characteristics that raise per-unit commuting costs, and θ_{pri} is a technology-specific shifter that captures differences in the marginal operating cost of private transit. The parameter β is the elasticity of congestion, while γ governs the elasticity with which private infrastructure reduces commuting costs.

Similarly, for public transit, traveling on edge (j, k) uses a constant amount of resources given

by

$$T_{jk}^{\text{pub}} = \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} L_{jk}^{\text{pub}}, \quad (4)$$

where parameter $\delta_{jk}^{\tau, \text{pub}}$ captures link-specific public-transit frictions, while θ_{pub} captures the marginal operating cost of the public technology.

We assume that public transit has no congestion but has capacity constraints. Public ridership must satisfy

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}, \quad \forall (j, k) \in \mathcal{E}, \quad (5)$$

where $\kappa_{jk}^{\text{pub}} > 0$ converts public infrastructure into public passenger capacity.² Infrastructure does not lower the per-rider public operating cost in (4); it expands the set of feasible public flows.

The total commuting resource cost on a link is the sum of mode-specific resource costs:

$$T_{jk} = T_{jk}^{\text{pub}} + T_{jk}^{\text{pri}}. \quad (6)$$

This additive formulation is central to preserving convexity, which will allow a much more efficient computational solution to the model. We will explain this in detail in the next section.

Building costs. We assume that building investment requires the use of an exogenous resource K that is in fixed aggregate supply in the economy. This resource is the government's budget. The cost of building infrastructure varies across links and technologies, reflecting geographic heterogeneity. Building one unit of public infrastructure on link jk requires $\delta_{jk}^{I, \text{pub}}$ units of K , while building one unit of private infrastructure requires $\delta_{jk}^{I, \text{pri}}$ units. The link-specific costs $\delta_{jk}^{I, \text{pub}}$ and $\delta_{jk}^{I, \text{pri}}$ capture how geographic features—such as terrain, geology, existing urban density, and land acquisition costs—differentially affect the cost of building each type of infrastructure. Aggregating across all links in the economy, we obtain the following resource constraint:

$$\sum_{(j,k) \in \mathcal{E}} \left[\delta_{jk}^{I, \text{pub}} I_{jk}^{\text{pub}} + \delta_{jk}^{I, \text{pri}} I_{jk}^{\text{pri}} \right] \leq K. \quad (7)$$

Flow constraint. At each location, effective labor demand must be feasible given local residents and incoming commuters. Because commuting consumes labor resources, outgoing flows enter the constraint together with their mode-specific resource costs:

$$\underbrace{L_j^L}_{\text{Local workers}} + \underbrace{\sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{jk}^m}_{\text{Outgoing commuters}} + \underbrace{\sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} T_{jk}^m}_{\text{Commuting resource costs}} \leq \underbrace{L_j^R}_{\text{Residents}} + \underbrace{\sum_{i \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{ij}^m}_{\text{Incoming commuters}}. \quad (8)$$

The right-hand side is the amount of labor available at node j : residents plus commuters arriving from neighboring nodes. The left-hand side is the amount of labor used at node j : local

²This is the minimal convex way to make the public network endogenous while preserving the assumption that public transit does not create congestion.

employment, commuters sent to neighboring nodes, and labor-equivalent commuting losses.

Preferences. Workers derive utility from consumption of a homogeneous freely-traded final consumption good C and a non-traded good H , i.e., residential land or housing. Utility of an individual worker who consumes c units of the final good and h units of land is

$$U(c_j, h_j) = c_j^\alpha h_j^{1-\alpha}.$$

Production and land. There are only two goods in this economy: a freely-traded final consumption good and a non-traded good. Each location produces the final good with a Cobb-Douglas technology combining productivity, labor, and commercial land. That is,

$$C_j = A_j L_j^{L^\psi} H_j^{c^{1-\psi}}$$

Regarding the non-traded good, land, we assume that each location has a fixed supply of residential and commercial land that we denote by H_j and H_j^c , respectively.

2.1 Planner's Problem

The planner maximizes welfare by choosing consumption and labor allocations across locations, commuting flows across edges, and infrastructure investments in each link of the transit network. Formally, the planner solves

$$\begin{aligned} \max \quad & u \\ & \{c_j, h_j, L_j^L, L_j^R\}_{j \in \mathcal{J}}, \\ & \{L_{jk}^m\}_{(j,k) \in \mathcal{E}, m \in \mathcal{M}}, \\ & \{T_{jk}^{\text{pub}}, T_{jk}^{\text{pri}}\}_{(j,k) \in \mathcal{E}} \end{aligned}$$

subject to the constraints described below.

Spatial equilibrium. The planner guarantees every resident at least utility u , so that no location offers strictly lower welfare:

$$u \leq U(c_j, h_j) \quad \forall j \in \mathcal{J}.$$

Balanced commuting flows. At each location, the supply of workers—residents plus incoming commuters—must cover demand—local workers plus outgoing commuters net of commuting costs:

$$\underbrace{L_j^L}_{\text{Local workers}} + \underbrace{\sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{jk}^m}_{\text{Outgoing commuters}} + \underbrace{\sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} T_{jk}^m}_{\text{Commuting resource costs}} \leq \underbrace{L_j^R}_{\text{Residents}} + \underbrace{\sum_{i \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{ij}^m}_{\text{Incoming commuters}}.$$

Commuting costs. The loss of resources on edge (j, k) depends on the modal composition of infrastructure, commuter volume, and total capacity:

$$T_{jk}^{\text{pri}} = \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}, \quad T_{jk}^{\text{pub}} = \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} L_{jk}^{\text{pub}}.$$

Public capacity. For every edge (j, k) , public ridership is bounded by public passenger capacity:

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}.$$

Population, goods, and land market clearing. Aggregate population is fixed at $\bar{L}$, aggregate final-good consumption cannot exceed aggregate production, and housing consumption is bounded by local residential land:

$$\sum_{j \in \mathcal{J}} L_j^R = \bar{L}, \quad \sum_{j \in \mathcal{J}} c_j L_j^R \leq \sum_{j \in \mathcal{J}} C_j, \quad h_j L_j^R \leq H_j \quad \forall j \in \mathcal{J}.$$

Network-building constraint. Total infrastructure spending is bounded by the aggregate budget K , and investment on each edge must respect pre-existing capacity bounds:

$$\sum_{(j,k) \in \mathcal{E}} \left[\delta_{jk}^{I, \text{pub}} I_{jk}^{\text{pub}} + \delta_{jk}^{I, \text{pri}} I_{jk}^{\text{pri}} \right] \leq K,$$

$$\underline{I}_{jk}^{\text{pub}} \leq I_{jk}^{\text{pub}} \leq \bar{I}_{jk}^{\text{pub}}, \quad \underline{I}_{jk}^{\text{pri}} \leq I_{jk}^{\text{pri}} \leq \bar{I}_{jk}^{\text{pri}} \quad \forall (j, k) \in \mathcal{E}.$$

Non-negativity. All allocation variables are non-negative:

$$c_j, h_j, L_j^R, L_j^L \geq 0, \quad L_{jk}^{\text{pub}}, L_{jk}^{\text{pri}} \geq 0, \quad I_{jk}^{\text{pub}}, I_{jk}^{\text{pri}} \geq 0.$$

The planner's problem can be thought of as a maximization problem in steps, where the planner first chooses i) the allocations of consumption, residents, and labor in each location, then ii) commuting flows across all location pairs, and finally iii) the optimal mix of public-private transit infrastructure investments along the network.

2.2 Optimal commuting flows

We now turn to analyze the main outcomes and economic mechanisms of the model. The central implication is that the optimal network is governed by comparative advantage across transit technologies. Public transit is attractive on links where its low operating cost and high passenger capacity compensate for the infrastructure resources required to build it. Private transit is attractive where its lower construction cost dominates, especially on low-volume links where congestion costs remain small. Thus, spatial variation in the optimal public-private network is driven by the interaction between operating costs, building costs, and equilibrium commuting volumes.

As in [Fajgelbaum and Schaal \(2020\)](#), the optimal flow allocation can be represented through a potential field. The multipliers on the local balanced-flow constraints determine the direction and intensity of commuting flows. Let P_j^F denote the multiplier on the balanced-flow constraint at node j , and let P^L denote the shadow value of aggregate labor. Define

$$v_j \equiv P_j^F + P^L.$$

The term $P_k^F - P_j^F$ is the marginal gain from moving one additional worker from node j toward node k . The term v_j is the value of one unit of labor-equivalent resources used in commuting from node j .

For public transit, the resource cost is linear:

$$T_{jk}^{\text{pub}} = \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} L_{jk}^{\text{pub}}.$$

Let η_{jk}^{pub} be the multiplier on the public-capacity constraint

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}.$$

The first-order condition for public flow is the no-arbitrage condition

$$P_k^F - P_j^F \leq (P_j^F + P^L) \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} + \eta_{jk}^{\text{pub}}, \quad (9)$$

with equality if $L_{jk}^{\text{pub}} > 0$. Thus the planner sends workers through public transit only when the value of moving a worker from j to k covers both the operating cost of public transit and the shadow cost of scarce public capacity.

Unlike private transit, public transit has constant marginal operating cost. Therefore (9) does not deliver an invertible formula for L_{jk}^{pub} . Instead, it determines whether public transit is active on the link. The level of public flow is pinned down jointly by network flow balance and the public-capacity constraint.

For private transit, the resource cost is

$$T_{jk}^{\text{pri}} = \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}.$$

The social marginal resource cost of private commuting is

$$MC_{jk}^{\text{pri}} = (1 + \beta) \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}. \quad (10)$$

The private-flow no-arbitrage condition is

$$P_k^F - P_j^F \leq (P_j^F + P^L) (1 + \beta) \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma},$$

with equality if $L_{jk}^{\text{pri}} > 0$. Since the private marginal cost is strictly increasing in private flow

when $\beta > 0$, this condition can be inverted:

$$L_{jk}^{\text{pri}} = \left[\frac{1}{1 + \beta} \frac{\left(I_{jk}^{\text{pri}}\right)^\gamma}{\delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}}} \frac{\max\{P_k^F - P_j^F, 0\}}{P_j^F + P^L} \right]^{1/\beta}. \quad (11)$$

Private flows therefore increase with the destination potential $P_k^F - P_j^F$ and with private infrastructure, and decrease with operating costs and congestion.

2.3 Optimal transit network

Given mode-specific commuting flows, the planner chooses public and private infrastructure. Let P^K be the multiplier on the network-building constraint.

Public infrastructure. Public infrastructure creates public passenger capacity. It does not directly reduce the per-rider operating cost in (4). Let η_{jk}^{pub} denote the multiplier on the public-capacity constraint

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}.$$

For an interior public-infrastructure choice, the first-order condition is

$$\underbrace{\eta_{jk}^{\text{pub}} \kappa_{jk}^{\text{pub}}}_{\text{Marginal benefit of public capacity}} = \underbrace{P^K \delta_{jk}^{I, \text{pub}}}_{\text{Marginal cost of public infrastructure}}. \quad (12)$$

This condition states that the marginal benefit of the additional public flow allowed by this capacity must be equal to the opportunity cost of using the scarce network-building resource on that edge.

Combining (9) and (12), an active public edge with interior infrastructure satisfies

$$P_k^F - P_j^F = (P_j^F + P^L) \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} + \frac{P^K \delta_{jk}^{I, \text{pub}}}{\kappa_{jk}^{\text{pub}}}. \quad (13)$$

The first term on the right-hand side is the public operating cost of carrying one more rider, valued in units of the planner's objective. The second term is the infrastructure cost of creating one more unit of public passenger capacity. Thus public transit is used on links where the value of moving workers across the link is large enough to cover both operating costs and the infrastructure required to serve them. Further, if public infrastructure has no lower bound and the upper bound is slack, then public capacity binds on every active public edge:

$$L_{jk}^{\text{pub}} = \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}. \quad (14)$$

Otherwise, the planner could reduce I_{jk}^{pub} without changing public flows, freeing resources for

other links. Hence, in this model public investment is proportional to the public flow it serves:

$$I_{jk}^{\text{pub}} = \frac{L_{jk}^{\text{pub}}}{\kappa_{jk}^{\text{pub}}}.$$

Private infrastructure. Private infrastructure works differently. It does not impose a hard capacity constraint; instead, it reduces the congestion cost of private commuting. For an interior choice of private infrastructure, the first-order condition is

$$\underbrace{(P_j^F + P^L)\gamma\delta_{jk}^{\tau,\text{pri}}\theta_{\text{pri}}\left(L_{jk}^{\text{pri}}\right)^{1+\beta}\left(I_{jk}^{\text{pri}}\right)^{-\gamma-1}}_{\text{Marginal reduction in private commuting costs}} = \underbrace{P^K\delta_{jk}^{I,\text{pri}}}_{\text{Marginal cost of private infrastructure}}. \quad (15)$$

The planner builds private infrastructure up to the point where the reduction in congestion costs equals the opportunity cost of the network-building resource. The benefit of private infrastructure is larger on links with more private flow, worse private commuting frictions, and a higher value of labor resources at the origin. It is smaller when private infrastructure is expensive to build or when the aggregate infrastructure budget is tight.

Solving (15) gives

$$I_{jk}^{\text{pri},*} = \left[\frac{\gamma(P_j^F + P^L)\delta_{jk}^{\tau,\text{pri}}\theta_{\text{pri}}\left(L_{jk}^{\text{pri}}\right)^{1+\beta}}{P^K\delta_{jk}^{I,\text{pri}}} \right]^{1/(1+\gamma)}. \quad (16)$$

This is the mode-specific analogue of the optimal infrastructure condition in [Fajgelbaum and Schaal \(2020\)](#). Private investment rises with private flow and with the value of reducing commuting losses, and falls with construction costs and the shadow price of the network budget. Importantly, this expression makes explicit that optimal private infrastructure depends crucially on relative marginal operating and building costs.

2.4 Technology choice and the cost trade-off

The previous conditions characterize public and private infrastructure separately. We can combine them to recover the main economic trade-off behind the optimal network. Public transit has a low and constant operating cost, but carrying public riders requires public capacity. Private transit does not face a hard capacity constraint, but private riders generate congestion, so the marginal cost of private service rises with the utilization of private infrastructure. The planner therefore compares two ways of producing mobility on each link: public capacity, which has a constant generalized unit cost, and private transit, whose optimized marginal cost depends on operating costs, construction costs, and congestion. The following proposition states the main trade-off.

Proposition 1 (Technology choice and cost trade-offs). Consider a link (j, k) with slack infras-

structure bounds. Suppose $\gamma > 0$ and $\beta \geq \gamma$. Define

$$V_j \equiv P_j^F + P^L,$$

$$A_{jk}^{\text{pub}} \equiv \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}}, \quad A_{jk}^{\text{pri}} \equiv \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}},$$

and let

$$D_{jk}^{\text{pub}} \equiv P^K \delta_{jk}^{I, \text{pub}}, \quad D_{jk}^{\text{pri}} \equiv P^K \delta_{jk}^{I, \text{pri}}$$

denote the shadow cost of building one unit of public and private infrastructure on link (j, k) . If technology-specific commuting costs are given by (3) and (4), then, on an active public edge, public capacity binds and the generalized unit cost of public service is

$$c_{jk}^{\text{pub}} = \underbrace{V_j A_{jk}^{\text{pub}}}_{\text{Operating cost}} + \underbrace{\frac{D_{jk}^{\text{pub}}}{\kappa_{jk}^{\text{pub}}}}_{\text{Capacity-building cost}}, \quad (17)$$

and the generalized unit cost of private service is

$$c_{jk}^{\text{pri}} (L_{jk}^{\text{pri}}) = (1 + \beta) \gamma^{-\gamma/(1+\gamma)} \left(V_j A_{jk}^{\text{pri}} \right)^{1/(1+\gamma)} \left(D_{jk}^{\text{pri}} \right)^{\gamma/(1+\gamma)} \left(L_{jk}^{\text{pri}} \right)^{(\beta-\gamma)/(1+\gamma)}. \quad (18)$$

If both modes are used on the same link, their generalized marginal costs are equal:

$$c_{jk}^{\text{pub}} = c_{jk}^{\text{pri}} (L_{jk}^{\text{pri}}). \quad (19)$$

Proof. See Appendix C.2. □

Proposition 1 characterizes the condition that determines the optimal mix of public and private infrastructure in the network. Equation (17) shows that public transit is attractive when it has a low operating cost, A_{jk}^{pub} , or when public capacity is cheap to create, $\delta_{jk}^{I, \text{pub}} / \kappa_{jk}^{\text{pub}}$. Equation (18) shows that private transit is attractive when private operating costs and private construction costs are low.

The main economic trade-off in the model is therefore that public transit is a low-operating-cost technology that requires costly capacity, while private transit's marginal cost rises with congestion but may compensate with low investment costs. The planner thus assigns flows and infrastructure so that, on links where both modes are active, the generalized marginal cost of public capacity equals the optimized generalized marginal cost of private congestible service.

2.5 Convexity and multiplicity

The planner's problem is a large-scale network design problem. In principle, this problem may have many local optima, since infrastructure choices on one edge affect the returns to infrastructure on other edges through equilibrium commuting flows. Convexity is therefore

important for both economic and computational reasons. Economically, it guarantees that the counterfactual network we compute is a global optimum rather than a local improvement around a particular initial guess. Computationally, it implies that the first-order conditions are sufficient and that different numerical solutions, if they arise, can be interpreted as either numerical error or as multiple global optima rather than as different local optima. Thus, convexity guarantees uniqueness of the counterfactual welfare level, while strict convexity would additionally guarantee uniqueness of the allocation.

We now state the condition under which our mode-specific formulation is convex.

Proposition 2 (Convexity of the planner's problem). Suppose that private commuting costs are given by

$$T_{jk}^{\text{pri}} = \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma},$$

public commuting costs are given by

$$T_{jk}^{\text{pub}} = \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} L_{jk}^{\text{pub}},$$

and public infrastructure enters through the linear capacity constraint

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}.$$

If $\beta \geq \gamma \geq 0$, then the mode-specific transport block is convex. Consequently, the full planner's problem is convex, up to the standard homothetic aggregate-variable transformation in the mobile-labor case,³ because all other constraints in the problem are linear or convex and the objective is linear in u . If $\beta = \gamma > 0$, the private transport block is convex but not strictly or strongly convex.

Proof. See Appendix C.3. □

Note that the relevant object is the total commuting resource cost that enters the balanced-flow constraint. For the congestible private technology, suppressing edge subscripts, the link-level resource cost is

$$T^{\text{pri}}(L, I) = AL^{1+\beta}I^{-\gamma}, \quad A > 0. \quad (20)$$

Intuitively, the parameter γ governs the returns to infrastructure, while β governs congestion. If infrastructure is too effective relative to congestion, then the transport technology has increasing returns: more infrastructure lowers costs, lower costs attract more flows, and more flows increase the incentive to invest in the same link. The condition $\beta \geq \gamma$ rules out this increasing-returns force by requiring congestion to be strong enough relative to the returns to infrastructure. The public technology, on the other hand, is convex by construction. Public op-

³By aggregate-variable representation, we mean rewriting individual consumption and housing choices as aggregate objects, for example $X_j = c_j L_j^R$ and $R_j = h_j L_j^R$, and using the perspective form implied by homothetic preferences. This removes bilinear terms such as $c_j L_j^R$ and $h_j L_j^R$ from the resource constraints. The per-capita notation in the text is expositional; the computational formulation uses the aggregate-variable version.

erating costs are linear in public ridership, and public infrastructure enters through the linear capacity constraint.

The full planner's problem inherits the convexity of the transport block. The objective is linear in the welfare level u . The population constraint, network-building constraint, public-capacity constraint, infrastructure bounds, and non-negativity constraints are linear. The production technology and the utility constraint are concave-side feasibility constraints under an aggregate-variable representation.

The boundary case $\beta = \gamma$ deserves special attention. In that case,

$$T^{\text{pri}}(L, I) = AL^{1+\beta}I^{-\beta}.$$

This function is homogeneous of degree one:

$$T^{\text{pri}}(tL, tI) = tT^{\text{pri}}(L, I) \quad \text{for all } t > 0.$$

Hence the function is linear along rays $(L, I) \mapsto (tL, tI)$. This implies that the private transport block is not strictly convex. As a result, different allocations of private flow and private infrastructure can deliver the same congestion ratio and the same welfare value.

Finally, if $\gamma > \beta$, the private transport block is non-convex. This is the increasing-returns case emphasized by [Fajgelbaum and Schaal \(2020\)](#). In that case, the first-order conditions are no longer sufficient for a global optimum, and the computed network may depend on the initial condition of the solver.

As we show in the next sections, we calibrate these parameters using detailed network data and obtain $\beta > \gamma$; thus, the baseline equilibrium and counterfactuals guarantee a unique global solution.

3 ILLUSTRATIVE EXAMPLE IN A STYLIZED CITY

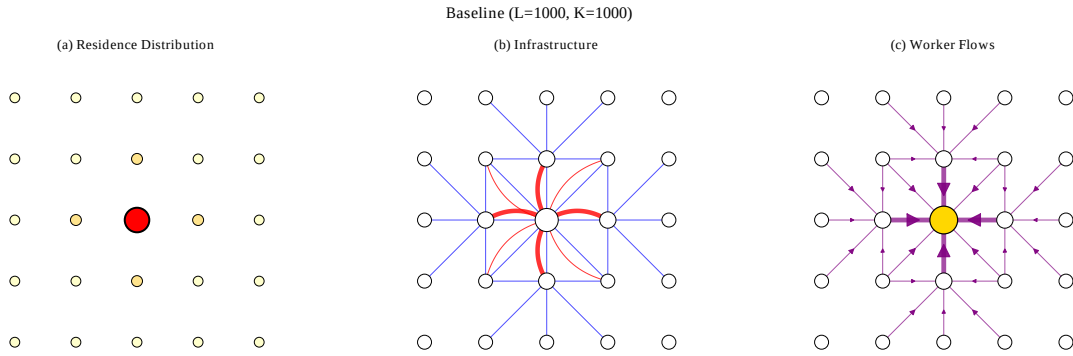
To build intuition and understand the key insights from the model, consider a stylized city that consists of a simple grid of 5-by-5 locations arranged on a two-dimensional plane. In this city, all nodes are equally productive except for the center node, which is twice as productive. All locations have the same endowment of the non-traded good, land. The size of the city is $\bar{L} = 1,000$ labor units. Locations are connected via a network in which each node has up to 8 neighbors (horizontal, vertical, and diagonal connections). The planner allocates a budget $K = 1,000$ across edges, choosing both the level of infrastructure investment and the share devoted to public versus private transit on each link. The baseline parameterization in this economy is

- Transport congestion parameters: $\beta = 0.8$ and $\gamma = 0.4$.
- Transport technology parameters: $\theta_{\text{pub}} = 0.1$ and $\theta_{\text{pri}} = 0.5$.
- Infrastructure costs: $\delta_I^{\text{pub}} = 5 \cdot \delta_{jk}$, and $\delta_I^{\text{pri}} = 1 \cdot \delta_{jk}$.

Here δ_{jk} is a Euclidean distance matrix normalized to 1 for vertical and horizontal edges. To build intuition, we set the marginal cost parameter of the private technology to be five times larger than that of the public technology and, conversely, the fixed cost of building public technology to be five times larger than that of the private technology.

Figure 1 illustrates the baseline equilibrium of the model in this environment, which helps clarify the main economic mechanisms at work. Although all locations are symmetric in terms of land endowments and network connectivity, the higher productivity of the central node induces a strong spatial reallocation of both residents and infrastructure. Population concentrates disproportionately at the center and, to a lesser extent, in nearby nodes, as shown in panel (a), reflecting the trade-off between commuting costs and access to higher productivity. The planner responds endogenously by concentrating public investment on the central corridors, while private links form a thinner access layer through the inner grid and toward the outer ring (panel (b)). As a result, commuting flows exhibit a pronounced radial pattern toward the central node, with the largest flows on the final approaches to the center (panel (c)). Together, these panels highlight how small exogenous productivity differences can generate large endogenous spatial asymmetries in population density, infrastructure provision, and commuting patterns.

FIGURE 1. EQUILIBRIUM OUTPUT: BASELINE EXAMPLE



Note: The figure illustrates the baseline equilibrium in a stylized 5-by-5 spatial economy. Nodes represent locations and edges denote commuting links between neighboring locations (horizontal, vertical, and diagonal). In panel (a), node size and color intensity capture the equilibrium residential distribution, with darker and larger nodes indicating higher population density. Panel (b) displays the optimal allocation of infrastructure investment across links. Red lines represent public infrastructure; blue lines represent private infrastructure. Thicker and darker edges correspond to higher total investment levels. Panel (c) shows equilibrium commuting flows, where arrow direction indicates the direction of worker movements and arrow thickness is proportional to the volume of commuters. The central node is twice as productive as all other locations. Total labor supply is fixed at $L = 1000$ and total infrastructure spending is constrained by $K = 1000$.

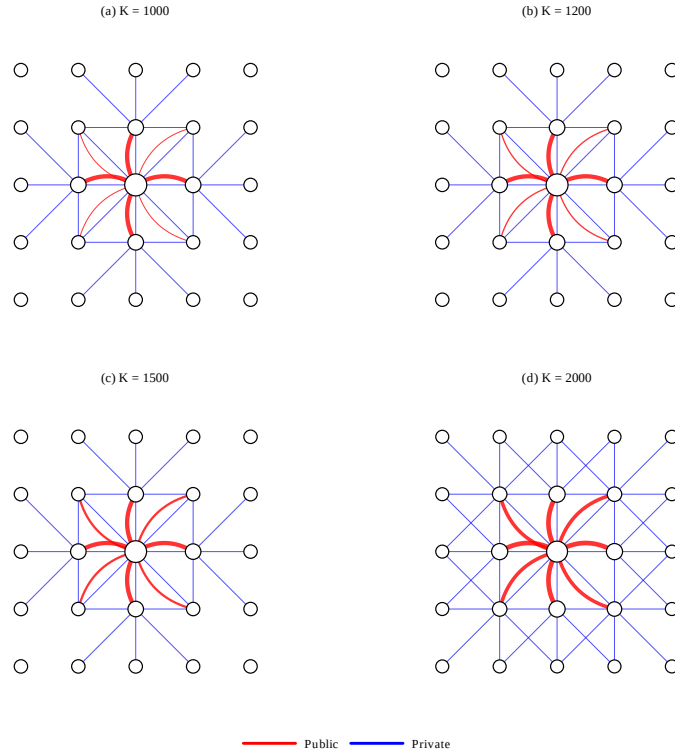
3.1 Comparative Statics

In this subsection, we examine how the network is designed when i) we increase the planner's budget, ii) we change the marginal cost parameters, or iii) we change the marginal building-cost parameters. In the following exercises, we describe the intuition behind each of these

scenarios.

Increasing budget K . How does the network change when we increase the planner's budget? In particular, we multiply the baseline budget by factors of 1.2, 1.5, and 2. Figure 2 shows the optimal transit network for different values of K . The baseline case is shown in panel (a). First, public infrastructure (red curves) is concentrated on the links connecting the center to its immediate neighbors, reflecting high commuting demand toward the productive center. As K increases, these public links expand mainly on the intensive margin: the red curves around the CBD become thicker, while public investment remains concentrated on the highest-flow corridors. Second, private infrastructure (blue lines) provides a low-cost connectivity layer around the public core. For $K = 1000, 1200$, and 1500 , this layer has a sparse star-and-inner-grid footprint that channels access toward the center. At $K = 2000$, private infrastructure expands across many more horizontal, vertical, and diagonal links in the outer grid, filling out the network where the fixed cost of public transit is not justified. Thus, additional resources do not lead the planner to build public transit everywhere; they deepen public capacity near the center and use private transit to extend the extensive margin of connectivity.

FIGURE 2. OPTIMAL NETWORK FOR DIFFERENT BUDGETS K

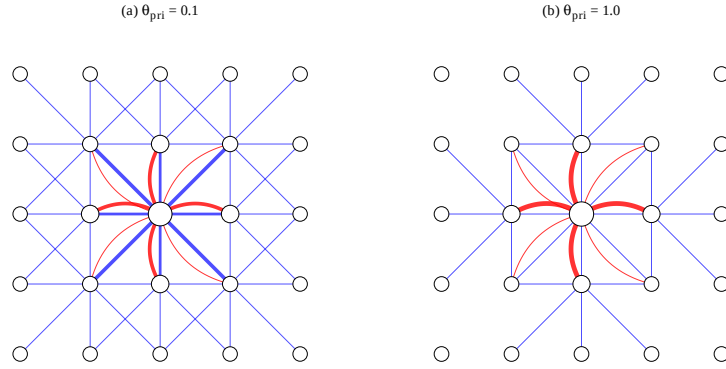


Note: The figure shows the optimal infrastructure allocation for a 5-by-5 grid city with a single CBD at the center. Red curved lines represent public transit infrastructure; blue straight lines represent private transit infrastructure. Line width is proportional to infrastructure investment, normalized globally across all panels to enable direct comparison. Node size reflects the number of residents at each location. Baseline calibration: $\bar{L} = 1,000$, $\beta = 0.8$ (congestion), $\gamma = 0.4$ (infrastructure elasticity), $\alpha = 0.7$ (consumption share), $\theta_{\text{pub}} = 0.1$, $\theta_{\text{pri}} = 0.5$, $\delta_{\text{pub}}^I = 5\delta$, $\delta_{\text{pri}}^I = \delta$.

Increasing marginal costs θ . How does the network change when the private technology

becomes relatively cheaper or more expensive to operate? Relative to the baseline scenario where $\theta_{\text{pri}} = 0.5$, here we compare $\theta_{\text{pri}} = 0.1$ and $\theta_{\text{pri}} = 1.0$. Figure 3 shows the networks under these parameters in panels (a) and (b), respectively. When private transit is cheap to operate ($\theta_{\text{pri}} = 0.1$, panel (a)), private infrastructure expands across much of the grid and becomes thick on some central and diagonal links. Public transit remains important on the highest-demand corridors, but the planner can rely more heavily on private links to extend network coverage. When private transit becomes more expensive ($\theta_{\text{pri}} = 1.0$, panel (b)), the private footprint contracts back toward the inner grid and the central public corridors become relatively more important. In both cases, public transit is concentrated where flows are thickest, while private transit provides flexible connectivity where the construction cost advantage is valuable. This illustrates a key insight: the optimal technology mix depends not on absolute costs but on the relative efficiency of public versus private provision.

FIGURE 3. OPTIMAL NETWORK FOR DIFFERENT MARGINAL COSTS

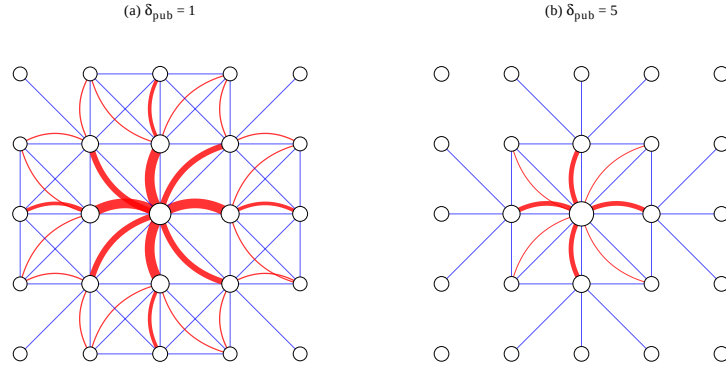


Note: The figure shows the optimal infrastructure allocation for a 5-by-5 grid city with a single CBD at the center. Red curved lines represent public transit infrastructure; blue straight lines represent private transit infrastructure. Line width is proportional to infrastructure investment, normalized globally across all panels to enable direct comparison. Node size reflects the number of residents at each location. Baseline calibration: $\bar{L} = 1,000$, $\beta = 0.8$ (congestion), $\gamma = 0.4$ (infrastructure elasticity), $\alpha = 0.7$ (consumption share), $\delta_{\text{pub}}^I = 5\delta$, $\delta_{\text{pri}}^I = \delta$.

Increasing building costs δ . How does the network change when the public technology becomes relatively cheaper or more expensive to build? Holding $\delta_{\text{pri}}^I = \delta$ fixed, Figure 4 compares a case where public infrastructure is as cheap to build as private infrastructure ($\delta_{\text{pub}}^I = \delta$) to the baseline case where public infrastructure is five times more expensive ($\delta_{\text{pub}}^I = 5\delta$). When public infrastructure is cheap to build (panel (a)), the planner builds a much denser mixed network: public transit is thick on central corridors and also appears on additional links around the inner grid and the outer edges, while private infrastructure creates broad connectivity across the city. When public infrastructure is more expensive (panel (b)), investment is much more concentrated: public transit remains on the central spokes, and private transit appears as a sparse access layer around the center and toward the outer ring. The planner therefore preserves public transit for the routes where its operating-cost advantage is most valuable and relies less on it when construction costs rise.

The key insight from this exercise illustrates the main trade-offs of the model: construction costs (δ) and operating costs (θ) interact to determine the optimal technology mix. Public transit is used where its low operating cost can justify the fixed cost of capacity, while private transit provides a lower-cost way to preserve connectivity on links where demand is not large enough to support public investment. The quantitative relevance of these mechanisms depends on the empirical magnitudes of the cost parameters and the structure of the observed network. We document these features for Mexico City in the next section before turning to calibration and counterfactuals.

FIGURE 4. OPTIMAL NETWORK FOR DIFFERENT BUILDING COSTS



Note: The figure shows the optimal infrastructure allocation for a 5-by-5 grid city with a single CBD at the center. Red curved lines represent public transit infrastructure; blue straight lines represent private transit infrastructure. Line width is proportional to infrastructure investment, normalized globally across all panels to enable direct comparison. Node size reflects the number of residents at each location. Baseline calibration: $\bar{L} = 1,000$, $\beta = 0.8$ (congestion), $\gamma = 0.4$ (infrastructure elasticity), $\alpha = 0.7$ (consumption share), $\theta_{pub} = 0.1$, $\theta_{pri} = 0.5$.

4 PRIVATE TRANSIT IN DEVELOPING CITIES: FACTS FROM MEXICO CITY

We now turn to Mexico City to document that the model’s central ingredients—the coexistence of private and public transit provision and the complementarity between modes—are quantitatively salient features of the data, and to motivate the calibration strategy of Section 5.

In developing cities, informal transit is not a transitional fix but a structural feature of urban mobility: privately operated minibuses and colectivos today provide between 50 and 100 percent of urban transit trips in many cities (Conwell, 2024), yet their interaction with public mass transit is poorly understood. Recent evidence shows these modes are far from independent—public investments reshape prices, waiting times, and route coverage through equilibrium responses of private operators (Björkegren et al., 2025; Conwell, 2024)—so studying them in isolation risks missing the most important channels through which infrastructure shapes welfare.

Mexico City is an ideal setting to address these questions. It combines one of the largest pub-

lic rail networks in the Western Hemisphere with an extensive web of informally operated minibuses, operates under binding fiscal constraints, and benefits from unusually rich micro-data on both modes (Mosqueda, 2026). Before documenting the set of facts that discipline the model’s calibration and motivate the counterfactual exercises in Section 6, we describe the data sources that underlie our analysis.

4.1 Data

We exploit a unique dataset that describes the universe of private and public transit routes and combine it with granular administrative data.

WhereIsMyTransport data. Public and privatized transport supply is described using a GTFS (General Transit Feed Specification) dataset produced by WhereIsMyTransport for 2018. A GTFS is a standardized dataset specification composed of a set of interrelated text files that provide a structured description of transit systems. These files define route identifiers, stop locations, trip schedules, service calendars, and operational attributes, allowing for a spatially explicit representation of the transport network that can be linked to geographic information systems. In addition to the usual formal public transport services such as subway and BRT, the WhereIsMyTransport GTFS feed incorporates private providers such as *microbuses* and *combis*, which are often underrepresented or absent in official transit datasets. For these services, the dataset reports route geographies, stop locations, operator or association identifiers, service frequencies, and weekly schedules. As a result, the GTFS feed provides a comprehensive description of both public and private transport supply in the metropolitan area, enabling consistent measurement of network structure and spatial coverage across modes.

TABLE 1—GTFS LINE CHARACTERISTICS BY TRANSIT TECHNOLOGY

Statistic	Public	Private
Lines	96	2,191
Stops per line	26.1	15.1
Line length (km)	15.5	13.1
Line speed (km/h)	24.8	15.8
Time between stops (min)	1.9	4.3
Average peak headway (min)	5.9	9.7
Average lowest scheduled headway (min)	5.6	8.2
Route-km share (%)	4.9	95.1

Note: Notes: The table summarizes the transit routes and schedules in the 2018 WhereIsMyTransport General Transit Feed Specification (GTFS) data for the Mexico City metropolitan area. A line is a distinct route pattern in the schedule data, so branches and opposite directions of the same named route may be counted separately. Private lines are informal colectivo and minibus services; public lines are formal rail, bus rapid transit, trolleybus, cable, and suburban rail services, following the same classification used in the calibration. Line length is measured along the mapped route geometry. Stops, length, speed, and time between stops are first computed at the scheduled-trip level and then averaged to the line level using scheduled service frequency as weights; table entries report averages across lines. Average peak headway is the average time between departures during weekday 06:00–10:00 service. Average lowest scheduled headway is the shortest scheduled time between departures observed for each line over the service week, averaged across lines. Route-km share is each technology’s share of total classified line kilometers.

Table 1 summarizes the resulting transit supply measures by ownership type. The private net-

work accounts for the overwhelming majority of route coverage, with 2,191 line patterns and 95 percent of classified route-kilometers, compared with 96 public line patterns. Public lines are faster on average and have shorter headways, while private lines are more spatially extensive and operate with fewer stops per line and longer times between stops. These differences highlight the central role of private transit in providing network coverage, despite its lower scheduled speeds and less frequent service relative to formal public transport.

INEGI data. We rely on two key datasets from Instituto Nacional de Estadística y Geografía (INEGI). First, the Encuesta Origen-Destino (EOD) 2017 for the Zona Metropolitana del Valle de México is a large-scale household travel survey that records commuting flow patterns within the metropolitan area. The EOD provides trip-level microdata, including trip origin and destination, departure and arrival times, trip purpose, sequence of transport modes used, including both public and private modes, monetary travel costs by mode, and walking time at the end of the trip. Second, we use data from the Economic Census 2019, which describes wages at the firm (and workplace) level for the universe of firms in Mexico.

Urban Planning Ministry data. Residential and commercial land use data come from the Urban Planning Ministries of Mexico City and the State of Mexico (Secretaría de Desarrollo Urbano e Infraestructura (SEIDUVI) and Secretaría de Desarrollo Urbano y Vivienda (SEDUVI)). These datasets describe land use at the tract, block, and residential/commercial unit levels.

4.2 Fact 1: Private Transit Dominates Network Coverage and Peripheral Connectivity

The first fact is about the scale of private transit relative to public transit. Private modes account for the large majority of route-kilometers and origin–destination pairs served in developing cities (Conwell, 2024; Kreindler et al., 2023), yet this is often overlooked in policy discussions that focus on formal infrastructure. Understanding the size of private transit is necessary for any welfare analysis of network design, since it determines the baseline connectivity that commuters face and how much public investment can improve access.

Table 2 shows the relative size of private and public transit in the Mexico City metropolitan area. Private transit accounts for 85 percent of total route-kilometers in the network, compared to just 15 percent for public mass transit—a ratio of nearly 6 to 1. In absolute terms, the private network covers 28,373 route-kilometers versus 5,002 for formal public transit (Appendix Figure 11). This coverage is spatially uneven: Appendix Figure 9 shows that formal public infrastructure—Metro, BRT, and commuter rail—is concentrated within about 15 km of the CBD, while private operators cover nearly all route-kilometers in peripheral districts, with many at or above 90 percent private coverage. Despite this large geographic footprint, private transit carries 53 percent of all transit trips while public transit carries 33 percent, reflecting that public infrastructure serves high-density, high-frequency corridors that concentrate demand. Among trips that use only one mode, 32 percent use only private transit versus 12 percent that

Metric	Private	Public
Share of total route km covered	0.85	0.15
Share of trips that use this transport	0.53	0.33
Share of trips that only use this transport	0.32	0.12
Avg share of trip time	0.35	0.19

TABLE 2—COMPARISON BETWEEN PRIVATE AND PUBLIC TRANSPORTATION IN USAGE AND ROUTE-KILOMETERS COVERED

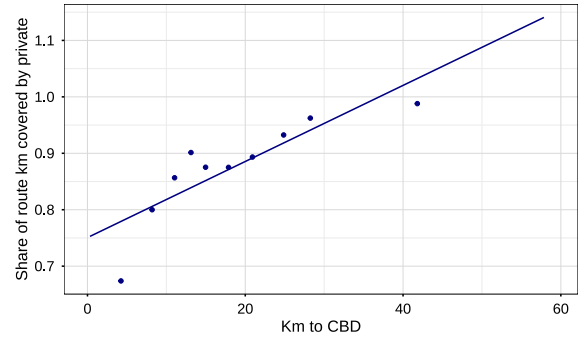


FIGURE 5. DISTANCE TO THE CBD AND SHARE OF KM COVERED BY PRIVATE TRANSPORTATION

use only public transit. On average, private transit accounts for 35 percent of total trip time, compared to 19 percent for public transit, showing that private modes make up a large share of the actual travel experience even on multimodal journeys.

The geographic distribution of coverage is closely tied to distance from the city center. Figure 5 shows a strong positive relationship between distance to the CBD and the share of route-kilometers covered by private transit: further from the center, private transit becomes increasingly dominant, filling gaps that public infrastructure does not reach. This gradient is confirmed in Appendix Figure 10, which breaks down total route-kilometers by distance band: private transit accounts for 68 percent of route-kilometers in the innermost ring (0–5 km), rising steadily to 100 percent beyond 35 km. This pattern is consistent with the model’s prediction that high fixed costs of public transit construction lead planners to concentrate public investment in high-density central corridors, leaving peripheral areas to private provision (Mosqueda, 2026).

4.3 Fact 2: Private and Public Transit Are Strongly Complementary

A key feature of Mexico City’s transit system is that public and private modes do not simply substitute for one another: they are tightly linked within individual commuting trips. This complementarity matters for understanding the welfare effects of transit policy, since any expansion of public infrastructure will affect not only the travelers who switch to it directly but also the private transit trips that feed into and out of it (Björkegren et al., 2025). How much commuters combine modes along a single journey also determines how operating cost differences between technologies show up in aggregate welfare—a key input to the model’s calibration. Tsivanidis (2019) documents substantial mode-combining behavior in Bogotá following the TransMilenio BRT expansion, where informal feeders absorbed much of the access demand created by new trunk lines. Similar patterns appear in the context of cable car expansions in Medellín (Khanna et al., 2024) and subway expansions in Mexico City (Zárate, 2024). Our data allow us to describe this complementarity more precisely and at finer spatial detail than prior work.

Metric	Value
Share of transit trips: public only	0.18
Share of transit trips: private + public (mixed)	0.33
P(uses private uses public)	0.65
Avg access+egress time on private (min)	37.34
Share metro stations with ≥ 1 combi stop (100 m)	0.81
Share metro stations with ≥ 2 combi stops (100 m)	0.64
Avg combi stops per metro station (100 m)	2.44

TABLE 3—COMPLEMENTARITY OF PUBLIC AND PRIVATE TRANSIT

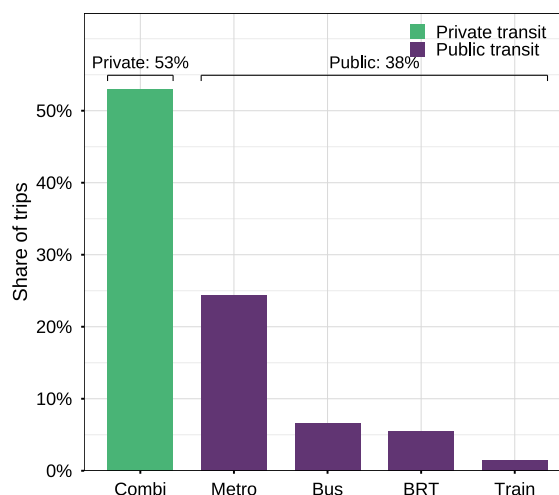


FIGURE 6. SHARE OF TRIPS BY TRANSPORTATION MODE

Table 3 documents several dimensions of this complementarity. Among all transit trips in the metropolitan area, 33 percent combine private and public services within a single journey (see also Appendix Figure 12, which shows the full breakdown of commutes into private-only, mixed, and public-only trips). Conditioning on trips that use public transit at least once, 65 percent also involve private transit, meaning that reaching public infrastructure almost always requires at least one private transit leg. The structure of these multimodal trips is informative: Appendix Figure 13 breaks each journey into access, trunk, and egress stages, showing that the most common pattern involves private access, public trunk, and private egress—consistent with combis playing a first-and-last-mile role. This is also visible at the mode-to-mode level: Appendix Figure 14 shows that combi-to-metro transitions account for 49.6 percent of all consecutive mode pairs within multimodal trips, confirming that private transit mainly serves as a feeder into public trunk infrastructure.

On average, commuters on mixed trips spend about 37 minutes on private transit for access and egress combined—a number that is similar to or exceeds the in-vehicle time on many public transit segments, meaning that private transit operating costs are incurred across nearly the full length of multimodal commutes. Appendix Figure 16 confirms that even on trips using public infrastructure, private transit takes up a large share of total travel time. Mixed trips are also the longest and most complex—averaging 96 minutes and 3.5 segments—while private-only trips are shorter (57 minutes) but the most expensive per trip (20.9 MXN), reflecting higher per-kilometer costs of combi service (Appendix Figure 17). These patterns are consistent with commuters combining modes to take advantage of the speed of public trunk services while relying on private transit for spatial coverage.

This first-and-last-mile role of private transit is also confirmed at the infrastructure level. Some 81 percent of metro stations have at least one combi stop within 100 meters, and 64 percent have two or more, with an average of 2.44 combi stops per metro station nearby (Appendix Figure 15). This co-location reflects how private feeder services have grown around public

infrastructure. Figure 6 adds to this picture: while combis account for the largest share of individual transit trips (53 percent), metro and BRT together account for 38 percent, and the overlap between these categories in individual journeys is large, showing that the two technologies serve complementary rather than competing roles.

These patterns have two important implications for the model. First, they support modeling public and private transit as jointly shaping network-level commuting costs, since individual journeys routinely combine private access and egress with public trunk segments. Second, they imply that the welfare gains from public transit investment come in part through their effect on the private transit trips that connect commuters to public infrastructure—a channel that would be missed entirely in a single-mode framework.

Together, these two facts show that private transit in Mexico City is not a marginal or residual phenomenon: it accounts for the large majority of network coverage (Fact 4.2) and plays a structural first-and-last-mile role in multimodal commuting (Fact 4.3). They also highlight the central tension that motivates the model: private transit fills gaps that public infrastructure cannot afford to serve, yet it does so at higher social cost and with greater congestion externalities than an unconstrained planner would choose (Mosqueda, 2026). Quantifying the welfare cost of this arrangement, and the gains from moving toward an optimal mix of public and private investment, is the task of the remainder of the paper.

5 CALIBRATION

5.1 Building a simplified network

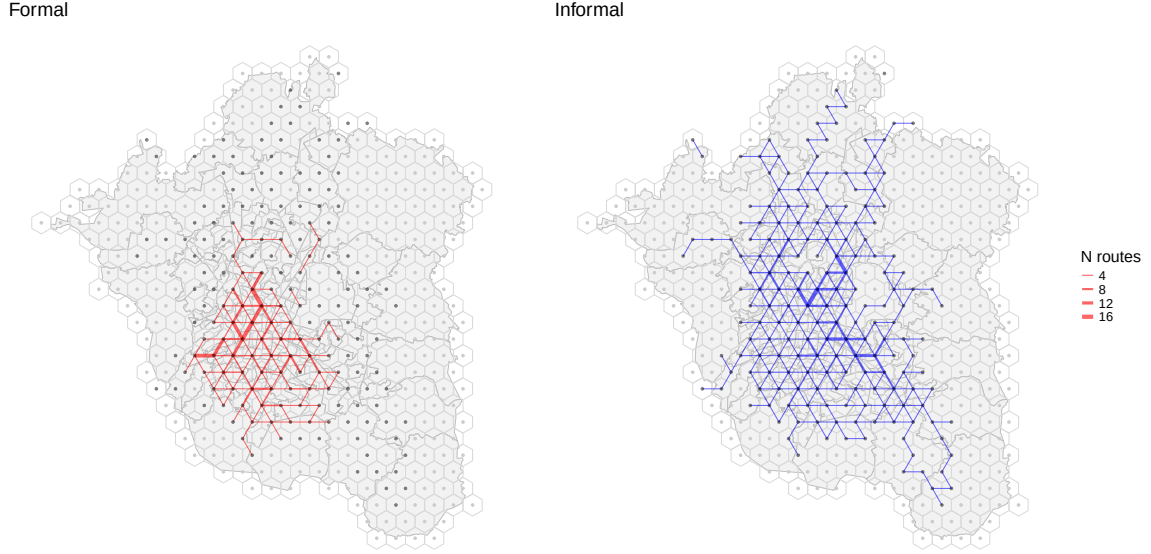
Transit systems are more complex to represent than standard road networks. GTFS data contain many overlapping routes across multiple modes, transfer points, direction-specific patterns, service-frequency differences, and mode/operator heterogeneity, all of which generate a very high-dimensional network object. A simplified representation is therefore useful: it preserves the broad spatial structure and baseline service patterns while keeping the state space manageable for computation.

We build a simplified transport network from GTFS route shapes by overlaying the metropolitan study area with a regular hexagonal grid of 5-km flat-to-flat hexagons. Figure 7 shows the simplified network by transit type. Each hexagon is treated as a node, located at its centroid. Hexagons are convenient because they provide a uniform spatial partition, avoid dependence on irregular administrative boundaries, and yield a relatively isotropic local neighborhood structure (each interior cell has the same number of immediate neighbors). This makes adjacency, distance calculations, and spatial aggregation of fundamentals more stable and transparent. The grid is clipped to the metropolitan boundary, and we compute an undirected geometric adjacency graph by linking hexagons that share a border. This geometric graph defines the set of feasible links in the model. The final simplified graph consists of 427 nodes and 1,187 edges.

GTFS information is then used to identify which neighboring hex-to-hex links are served by

transit and to construct baseline infrastructure measures. In broad terms, we map observed routes onto the hex grid, translate them into movements across adjacent cells, and then aggregate these movements into edge-level transit intensity measures. This gives us a simplified representation of existing service on the network, including route counts split into formal and informal services. This produces two related objects: i) an adjacency matrix over all neighboring hexagons—the set of links the model may invest in—and ii) GTFS-based edge attributes, including lower bounds for the observed public and private infrastructure.

FIGURE 7. SIMPLIFIED NETWORKS BY PUBLIC/PRIVATE TYPE



Note: The figure shows the simplified transit networks by public or private type. The city was discretized into symmetric hexagons, with centroids representing the nodes in the network. Neighboring nodes are connected if at least one transit line passes between them. The thickness of the lines represents the number of transit routes that connect two given nodes. Network data come from WhereIsMyTransport’s GTFS file.

5.2 Parameter calibration

Marginal costs of operation θ . To calibrate the marginal cost parameters θ_{pub} and θ_{priv} , we exploit the granular detail of the GTFS file. In particular, we use speed as a proxy for the efficiency of each technology and its social marginal cost. Speed reflects the quality of the infrastructure and the units that run through it, as well as the scope for potential congestion. Using the GTFS, we compute the speed of all trips recorded for each individual transit line, and we take the average speed of public and private lines. We normalize $\theta_{\text{pub}} = 1$ because these parameters are shifters in our commuting cost specification and only their relative levels matter. Taking the ratio of average speeds across public and private lines yields $\theta_{\text{priv}} = 1.64$. This suggests that minibuses are 64% more costly to operate than public options such as the subway, BRT, and train.

Building costs δ . To calibrate the construction parameters δ_{pub} and δ_{priv} , we follow [Fajgelbaum and Schaal \(2020\)](#) and use the government budget constraint (7) to recover the δ pa-

parameters that are consistent with the existing (observed) infrastructure. First, we define the infrastructure measures I_{jk}^{pri} and I_{jk}^{pub} to be the total number of private and public transit lines, respectively, connecting any two nodes in the network.⁴ Second, we assume building costs are constant scalars that vary only across public and private modes, but are otherwise constant across links. Third, because only the relative levels of δ_{pub} and δ_{priv} matter, we normalize $\delta_{\text{pub}} = 1$ and recover the δ_{priv} that is consistent with a given budget K . Given the levels of our infrastructure measures, we set $K = 2,000$ and recover $\delta_{\text{priv}} = 0.1$. This suggests that building private infrastructure is 10 times cheaper than building public infrastructure.

Elasticity of congestion. We take the value of the elasticity of congestion from Mosqueda (2026). We set $\beta = 0.77$. In his paper, this elasticity corresponds to the elasticity of road speed with respect to an additional minibus on the road and is estimated using exogenous variation generated from the collapse of a major subway line. The interpretation of this elasticity maps directly to our interpretation of private transit infrastructure generating congestion.

Infrastructure elasticity. To calibrate the elasticity of private commuting costs with respect to infrastructure, γ , we use cross-sectional variation in EOD district-pair commuting flows and GTFS-based measures of private transit supply. For each directed district pair, we construct the flow of work trips that use private transit in at least one leg, and we measure private infrastructure using the density of private GTFS routes along plausible paths connecting the origin and destination districts. We then estimate a PPML gravity-style regression of private commuting flows on the log private infrastructure index, controlling for origin and destination fixed effects, distance, and public-transit infrastructure. The coefficient on private infrastructure, $\hat{b}_I$, maps into the model parameter through $\hat{\gamma} = \beta \hat{b}_I$. Using the externally calibrated value $\beta = 0.77$, the benchmark estimate $\hat{b}_I = 0.097$ implies $\hat{\gamma} = 0.075$. Across alternative specifications that vary the dependent variable, the private-infrastructure measure, and the set of controls, the estimates are generally positive, with implied values of γ ranging from roughly 0.05 to 0.11. The benchmark PPML estimate is positive but imprecise; estimates based on positive-flow log-linear specifications and alternative measures of private route availability are more statistically precise. We therefore interpret the exercise as disciplining the order of magnitude of γ , rather than as causal identification. Since the benchmark value is positive, small relative to β , and satisfies the convexity restriction $\gamma < \beta$, we use $\gamma = 0.075$ in the benchmark and report robustness over a grid of values below β .

Commuting cost shifters. We calibrate the link-specific commuting cost shifters $\delta_{jk}^{\tau, \text{pub}}$ and $\delta_{jk}^{\tau, \text{pri}}$ by inverting the commuting cost equations in the model. These parameters capture geographic and network frictions that make commuting more or less costly across links, holding fixed the technology-specific cost shifters and the congestion/infrastructure elasticities. In particular, using the EOD we observe commuting flows and travel times across origin-destination pairs, and using GTFS we measure the corresponding public and private transit infrastructure.

⁴Fajgelbaum and Schaal (2020) use road lanes as their infrastructure measure.

Given calibrated values of θ_{pub} , θ_{pri} , β , and γ , we choose $\delta_{jk}^{\tau, \text{pub}}$ and $\delta_{jk}^{\tau, \text{pri}}$ so that equations (4) and (3) exactly reproduce observed commuting resource costs. Thus, public shifters rationalize observed public travel times given public flows, while private shifters rationalize observed private travel times given private flows and private infrastructure. We first perform this inversion at the EOD district-pair level, where travel times and flows are observed, and then map the resulting shifters to the model’s network edges. The calibrated δ^τ objects therefore summarize the residual spatial heterogeneity in commuting costs that is not explained by mode technology, congestion, or measured infrastructure. The median values of these jk matrices are $\delta_{jk}^{\tau, \text{pri}} = 0.00026$, and $\delta_{jk}^{\tau, \text{pub}} = 0.13320$.

Public infrastructure capacity. To calibrate κ , which captures how many people in a given link can actually travel using one unit of public infrastructure, we target the *Encuesta Origen-Destino* aggregate share of commuters using public (17%) and private (83%) transit. We set $\kappa = 924$.

Land and productivity. To obtain a measure of H and H^c , we rely on land use data that show residential and commercial land available at the tract level. We then aggregate to the hexagon units that define our network. For productivity A , we compute average wages at the census tract level using data from the Economic Census, which reports wages at the firm level. We then aggregate into our hexagon geographic units and map average wages directly to productivity.

6 WELFARE EFFECTS OF TRANSIT EXPANSIONS

In this section, we study the gains from budget-feasible expansions of the transit system. In particular, given the current network and spatial equilibrium, we assess where it is optimal to invest in transit infrastructure and how these investments differ across transit technologies. We depart from the baseline equilibrium in which the public and private networks are the observed ones. In our calibration, we assume that the current network was built with budget K , and the counterfactuals presented here increase K by 50%.

To highlight and assess the role of the private technology, which has received limited attention in the urban transit interventions literature, we perform two counterfactuals: i) one where the planner considers only public transit infrastructure investments, and ii) another where the planner considers both public and private infrastructure. Comparing these two scenarios allows us to assess the potential additional welfare gains of investing in private transit relative to public transit alone.

Welfare gains from network expansions. Table 4 shows the welfare gains from increasing the baseline budget by 50% and optimally expanding the current network. Focusing on the welfare gain column, the first row shows a 1.33% gain from expanding the network when the planner implements only public transit investments, while the second row shows a 1.59% gain

TABLE 4—WELFARE GAINS OF TRANSIT EXPANSIONS

Scenario	Δ Welfare (%)	Public budget share (%)	Public infr. share (%)	Public flow share (%)
(1) Build public only	1.334	100.000	15.452	56.246
(2) Build public and private	1.594	97.448	14.976	39.027

Note: The counterfactuals are relative to the baseline, which is the equilibrium with the observed network with no new investment ($K = 0$). Both scenarios consider a 50% expansion of the government's budget K .

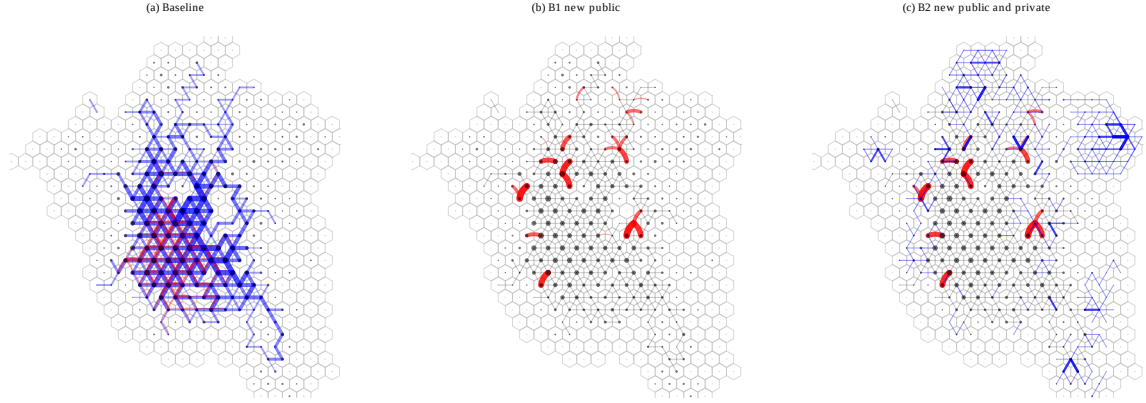


FIGURE 8. OPTIMAL TRANSIT EXPANSIONS

Note: The figure shows two counterfactual scenarios of expanding transit infrastructure relative to the current network. We consider a 50% increase in budget K . The B1 scenario shows the optimal expansion when the planner uses public transit exclusively. The B2 scenario shows the optimal expansion when the planner can use both public and private transit. Red lines indicate new public infrastructure and blue lines indicate new private infrastructure. Grey lines represent existing infrastructure. Node size represents the distribution of population by residence, with larger nodes indicating higher population density.

when the planner can invest in both technologies. The comparison suggests that private transit can deliver an extra marginal welfare gain of 0.26 percentage points, or roughly a 20% additional welfare gain. This magnitude is quantitatively important: governments in developing cities could benefit substantially by considering the joint design of the network, rather than the common recipe of expanding and evaluating public mass transit in isolation.

To put the magnitude of these gains in perspective, in the developing-country setting, [Graff \(2024\)](#) found that reorganizing Africa's entire road network would result in a welfare gain of 1.3%, and that optimally expanding it by a tenth would result in a 0.8% gain. Our results align overall with these magnitudes, suggesting they are economically meaningful.

Spatial effects. Where should the planner expand the network? Figure 8 shows the spatial distribution of counterfactual investments. The left panel shows the baseline public and private networks. The middle panel shows the optimal expansion when the planner can only add public infrastructure. The right panel shows the optimal expansion when the planner can expand both public and private transit. In all panels, grey lines represent existing infrastructure and node size represents the distribution of population by residence.

The public-only expansion in the middle panel is not simply a radial expansion around the city center. Instead, the planner places new public infrastructure in a small number of high-return

corridors where large commuting flows or productive destinations make the high fixed cost of public investment worthwhile. A first set of investments appears in the northwest, around the Tlalnepantla–Naucalpan–Atizapán corridor. These locations form a productive satellite-city area with limited integration into the existing mass-transit network. A second set of investments appears around the airport (East), which is another important employment and connectivity node. A third set appears toward Ecatepec in the northeast. Unlike the northwest corridor and the airport, this area is not among the most productive locations, but it attracts and generates very large commuting flows. The planner therefore invests there because improving access for a large mass of commuters generates large aggregate gains even when destination productivity is not especially high.

Comparing the public-only expansion with the mixed public–private expansion highlights the role of technology choice. When the planner can use both technologies, public investment becomes more selective: it is used on links where the combination of thick flows, productive destinations, and network centrality justifies the high fixed cost of public infrastructure. Private investment, by contrast, expands much more widely across the periphery. This reflects the lower cost and greater flexibility of the private technology. The private network is especially useful for extending coverage to peripheral areas, feeding commuters into the existing network, and connecting locations where demand is valuable but not large enough to justify public infrastructure. Thus, the optimal mixed network does not replace public transit with private transit everywhere. Rather, it assigns public infrastructure to the thickest and most central corridors, while using private transit to provide a lower-cost extensive margin of connectivity.

6.1 Discussion

Our model and quantitative results abstract from private automobile mode choice and therefore should be interpreted as a model of optimal transit network design, not of the full multi-modal urban transport equilibrium. This abstraction is deliberate. In Mexico City, the empirically central and under-studied margin for transit planning is the interaction between formal public trunk infrastructure and extensive private/informal feeder services, which jointly determine a large share of transit connectivity and commuting costs. Incorporating private cars would add important substitution and road-congestion channels, but would require a substantially richer model of mode choice, car ownership, and shared road capacity. We therefore view our results as characterizing the optimal allocation of transit technologies conditional on the broader automobile environment. To the extent that improved transit also draws travelers from cars, our estimates may understate the broader welfare gains from coordinated transit planning. The welfare results presented here can therefore be interpreted as a lower bound.

6.2 Robustness to alternative parameters

Appendix B reports sensitivity exercises that vary the infrastructure budget, the marginal operating cost of private transit, and the construction cost of private infrastructure. In each exercise,

we compare the same two counterfactuals as above: B1 allows only public-transit expansion, while B2 allows the planner to expand both public and private transit. Overall, the welfare patterns are stable across these perturbations. The mixed public–private counterfactual always delivers larger gains than the public-only counterfactual, and the magnitudes remain close to those in the main exercise.

Table 5 shows that welfare gains increase monotonically with the size of the budget expansion. Increasing ΔK from 10 to 75 percent raises the public-only welfare gain from 0.35 to 1.86 percent and the mixed public–private gain from 0.62 to 2.11 percent. The incremental gain from allowing private investment is positive at all budget levels and remains approximately stable, at around one-quarter of a percentage point. Thus, larger budgets raise welfare in both planning problems, but the role of private transit remains quantitatively similar: it provides an additional margin of network expansion beyond what can be achieved by public infrastructure alone.

The cost sensitivity exercises reinforce the model’s technology-choice mechanism. Table 6 increases θ_{pri} , making private transit more expensive to operate. As private operating costs rise, the welfare gain from the mixed counterfactual converges toward the public-only counterfactual: the incremental B2–B1 gain declines from 0.38 percentage points when $\theta_{\text{pri}} = 1.05$ to 0.04 percentage points when $\theta_{\text{pri}} = 10$. Hence, when private transit becomes sufficiently costly to operate, the additional value of allowing private investment becomes small. Table 7 varies the private construction-cost shifter. Although total welfare gains rise in this exercise because the calibrated budget also rises, the incremental gain from allowing private investment falls from 0.31 to 0.21 percentage points as private infrastructure becomes more expensive to build. This pattern is consistent with the same trade-off: private transit contributes most when it combines low fixed costs with acceptable marginal operating costs.

Taken together, the sensitivity analysis suggests that the main quantitative conclusion is not driven by a narrow parameter choice. The level of welfare gains changes with the budget and with private-transit cost parameters, but the qualitative ranking of counterfactuals is robust. Public investment remains the main source of gains on high-demand corridors, while allowing private investment adds a smaller but positive welfare gain by extending the network on links where public infrastructure is too costly relative to demand.

7 CONCLUSION

This paper studies how a budget-constrained planner should optimally allocate public and private transit technologies across an urban transportation network. The central trade-off is that mass transit technologies are fixed-cost intensive but have low marginal operating costs, while private transit requires little upfront investment but involves higher marginal costs and generates congestion externalities. Understanding how to optimally combine these technologies is particularly important in developing cities, where fiscal constraints limit the expansion of mass transit infrastructure and private transit plays a central role in providing commuting mobility.

We develop a quantitative spatial equilibrium framework in which a planner chooses commuting flows, spatial allocations, and the technology composition of transit infrastructure across the network. The model delivers a simple characterization of the optimal public-private technology mix: the optimal technology deployed on each link depends on the trade-off between relative marginal operating costs and infrastructure construction costs across technologies. Public transit is optimally deployed on high-demand corridors where its low operating costs can be amortized over large flows, while private transit provides baseline connectivity across the rest of the network where building mass transit infrastructure is too expensive relative to demand.

To bring the model to the data, we construct a new dataset that combines GTFS data describing the full universe of public and private transit routes with administrative microdata on commuting flows, wages, and land use for Mexico City. The GTFS data allow us to build a unified representation of the city's transit network, including informal and private routes that are typically missing from official datasets, and to map transit supply into a simplified network structure suitable for quantitative analysis. Using these data, we document two key facts that motivate the framework: private transit dominates network coverage and peripheral connectivity, and public and private transit are strongly complementary within individual commuting trips.

Quantitatively, we find that the welfare gains from transit expansions depend critically on whether private transit is incorporated into the planning problem. When public infrastructure is expanded without accounting for the private network, welfare gains are substantially smaller. Allowing the planner to condition public investment on the existing private network improves outcomes, while jointly planning public and private infrastructure yields the largest welfare gains. These results imply that a significant share of the benefits from transit investment arises not only from expanding public infrastructure itself, but from optimally integrating private transit into the network design.

Taken together, the results suggest that private transit in developing cities should not be viewed solely as an inefficient or transitional sector, but rather as a technology that can play an important role in the optimal transit system when governments face binding fiscal constraints. The optimal policy is therefore not to replace private transit everywhere with mass transit, but to jointly design both systems so that public infrastructure serves high-capacity corridors while private transit provides flexible connectivity across the rest of the city. More broadly, the paper highlights that infrastructure policy in developing cities should be understood as a problem of optimal technology allocation under fiscal constraints, rather than as a choice between formal and informal systems.

Finally, an important avenue for future research is to explore the distributional consequences of designing private and public networks. We show that optimal network expansions should improve connectivity between peripheral locations and central infrastructure, potentially benefiting low-income workers who reside in such peripheral locations. An analysis of the distributional impacts is therefore important for assessing the equity-efficiency trade-offs in the

design of transit networks.

REFERENCES

- ADAMOPOULOS, T., L. BRANDT, J. LEIGHT, AND D. RESTUCCIA (2022): “Misallocation, selection, and productivity: A quantitative analysis with panel data from China,” *Econometrica*, 90, 1261–1282.
- ALDER, S. (2025): “Chinese roads in India: The effect of transport infrastructure on economic development,” *Journal of International Economics*, 104140.
- ALLEN, T. AND C. ARKOLAKIS (2022): “The welfare effects of transportation infrastructure improvements,” *The Review of Economic Studies*, 89, 2911–2957.
- ALMAGRO, M., A. BARBIERI, J. C. CASTILLO, A. HICKOK, AND T. SALZ (2024): “Optimal Urban Transportation Policy: Evidence from Chicago,” Working paper.
- ASTURIAS, J., M. G. SANTANA, AND R. RAMOS (2014): *Misallocation, internal trade, and the role of transportation infrastructure*, CEMFI (Centro de Estudios Monetarios y Financieros) Madrid, Spain.
- BAUM-SNOW, N. (2007): “Did highways cause suburbanization?” *The quarterly journal of economics*, 122, 775–805.
- BERNOT, M., V. CASELLES, AND J.-M. MOREL (2009): *Optimal transportation networks: models and theory*, Springer.
- BJÖRKEGREN, D., A. DUHAUT, G. NAGPAL, AND N. TSIVANIDIS (2025): “Public and Private Transit: Evidence from Lagos,” Tech. rep., National Bureau of Economic Research.
- CHANDRA, A. AND E. THOMPSON (2000): “Does public infrastructure affect economic activity?: Evidence from the rural interstate highway system,” *Regional science and urban economics*, 30, 457–490.
- CONWELL, L. (2024): “Privatized Provision of Public Transit,” .
- DESMET, K. AND E. ROSSI-HANSBERG (2013): “Urban accounting and welfare,” *American Economic Review*, 103, 2296–2327.
- DONALDSON, D. (2018): “Railroads of the Raj: Estimating the impact of transportation infrastructure,” *American Economic Review*, 108, 899–934.
- DURANTON, G., P. M. MORROW, AND M. A. TURNER (2014): “Roads and Trade: Evidence from the US,” *Review of Economic Studies*, 81, 681–724.
- FABER, B. (2014): “Trade integration, market size, and industrialization: evidence from China’s National Trunk Highway System,” *Review of Economic Studies*, 81, 1046–1070.

- FAJGELBAUM, P. D., E. MORALES, J. C. SUÁREZ SERRATO, AND O. ZIDAR (2019): “State taxes and spatial misallocation,” *The Review of Economic Studies*, 86, 333–376.
- FAJGELBAUM, P. D. AND E. SCHAAL (2020): “Optimal Transport Networks in Spatial Equilibrium,” *Econometrica*, 88, 1411–1452.
- FELBERMAYR, G. J. AND A. TARASOV (2022): “Trade and the spatial distribution of transport infrastructure,” *Journal of Urban Economics*, 130, 103473.
- FERNALD, J. G. (1999): “Roads to prosperity? Assessing the link between public capital and productivity,” *American economic review*, 89, 619–638.
- GRAFF, T. (2024): “Spatial inefficiencies in Africa’s trade network,” *Journal of Development Economics*, 171, 103319.
- HSIEH, C.-T. AND P. J. KLENOW (2009): “Misallocation and manufacturing TFP in China and India,” *The Quarterly journal of economics*, 124, 1403–1448.
- HSIEH, C.-T. AND E. MORETTI (2019): “Housing constraints and spatial misallocation,” *American economic journal: macroeconomics*, 11, 1–39.
- KHANNA, G., C. MEDINA, A. NYSHADHAM, J. RAMOS, J. TAMAYO, AND Z. H. TIEW (2024): “Spatial Mobility, Economic Opportunity, and Crime,” Working paper; add link when available.
- KREINDLER, G., A. GADUH, T. GRAFF, R. HANNA, AND B. A. OLKEN (2023): “Optimal Public Transportation Networks: Evidence from the World’s Largest Bus Rapid Transit System in Jakarta,” Tech. rep., National Bureau of Economic Research.
- MOSQUEDA, J. (2026): “Equilibrium Commuting Costs: The Role of Private and Public Transit,” Job Market Paper; working paper.
- ROGERSON, R. AND D. RESTUCCIA (2004): “Policy distortions and aggregate productivity with heterogeneous plants,” in *2004 Meeting Papers*, Society for Economic Dynamics, 69.
- TARASOV, A. AND G. FELBERMAYR (2014): “Trade and the Spatial Distribution of Transport Infrastructure,” .
- TSIVANIDIS, N. (2019): “Evaluating the Aggregate and Distributional Impacts of Urban Transit Infrastructure: Evidence from Bogotá’s TransMilenio,” Working paper; various revisions through 2023.
- ZÁRATE, R. D. (2024): “Spatial Misallocation, Informality, and Transit Improvements: Evidence from Mexico City,” Working paper.

APPENDIX

APPENDIX CONTENTS

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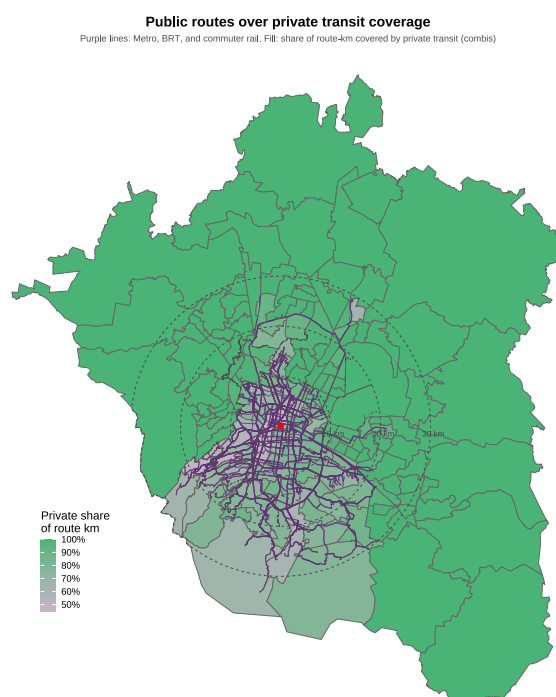
A TRANSIT NETWORK IN MEXICO

A.1 Private Transit and Network Coverage

This section documents the dominance of private transit operators (*combis*) in the ZMVM's transit network. We measure network coverage using total route-kilometers by mode and district.

Figure 9 displays a map of the ZMVM in which each district is shaded according to the share of route-kilometers covered by private transit (*combis*). Formal public transit routes—Metro, BRT (Metrobús), and commuter rail (Tren Suburbano)—are overlaid as purple lines, with concentric rings indicating distances of 10, 20, and 30 km from the CBD. Two patterns stand out. First, formal public transit infrastructure is concentrated within approximately 15 km of the CBD. Second, in the vast majority of districts—particularly those in the urban periphery—private operators account for nearly all route-kilometers, with many districts at or above 90% private coverage.

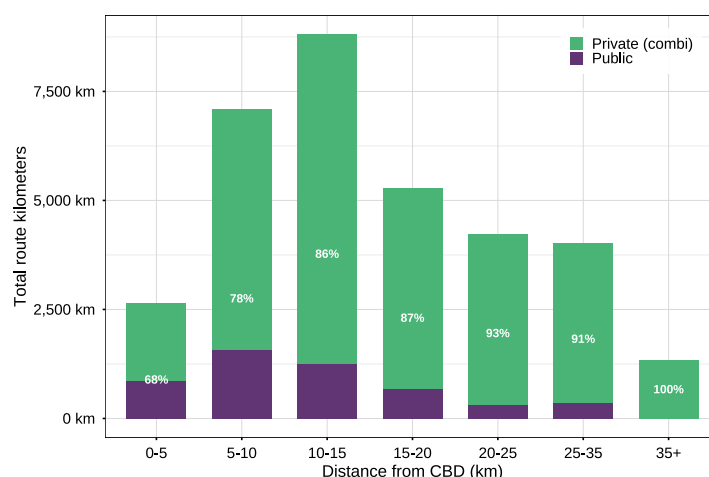
FIGURE 9. SPATIAL DISTRIBUTION OF PRIVATE TRANSIT COVERAGE



Notes: Each district is shaded by the share of route-kilometers served by private transit (*combis*). Purple lines indicate formal public transit routes (Metro, BRT, and commuter rail). Dashed circles mark 10, 20, and 30 km from the CBD (red diamond).

Figure 10 decomposes total route-kilometers by distance band from the CBD. Even in the innermost ring (0–5 km), private transit accounts for 68% of route-kilometers. This share rises monotonically with distance: from 78% in the 5–10 km band to 100% beyond 35 km. The gradient reflects the fact that formal public transit networks were built to serve central areas, leaving peripheral districts almost entirely dependent on informal private operators.

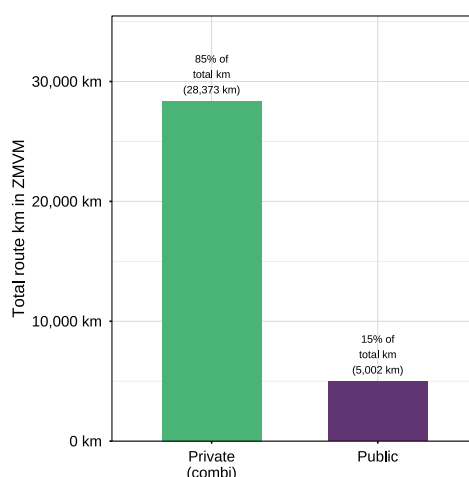
FIGURE 10. ROUTE-KILOMETERS BY DISTANCE FROM CBD



Notes: Each bar shows total route-kilometers by distance band from the CBD, decomposed into private (combi) and public transit. Percentages indicate the private share within each band.

Figure 11 summarizes aggregate network coverage across the entire ZMVM. Private transit accounts for 28,373 route-kilometers, or 85% of the total network. Formal public transit contributes 5,002 route-kilometers (15%). These figures underscore a key feature of the ZMVM's transit system: despite the visibility of large-scale public infrastructure such as the Metro, the backbone of the network in terms of spatial coverage is overwhelmingly composed of privately operated combi routes.

FIGURE 11. AGGREGATE NETWORK COVERAGE: PRIVATE VS. PUBLIC TRANSIT



Notes: The figure shows total route-kilometers in the ZMVM by operator type. Private transit (combis) accounts for 28,373 km (85%) and formal public transit for 5,002 km (15%).

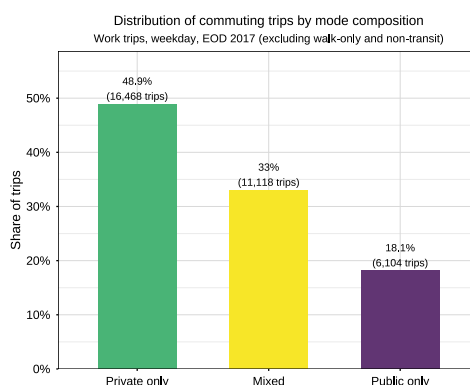
A.2 Additional Evidence on Public–Private Transit Complementarity

This section presents additional evidence supporting the complementarity between public and private transit documented in Section 4.3. We decompose commuting trips by mode composition, examine the spatial co-location of public and private stops, and characterize how trip

attributes vary across trip types.

Trip composition. Figure 12 classifies all weekday work commutes in the 2017 EOD (excluding walk-only and non-transit trips) into three categories: private only, mixed (combining public and private modes), and public only. Private-only trips account for 48.9 percent of commutes (16,468 trips), mixed trips for 33 percent (11,118 trips), and public-only trips for 18.1 percent (6,104 trips). The large share of mixed trips confirms that public and private transit are routinely combined within individual journeys rather than serving as substitutes.

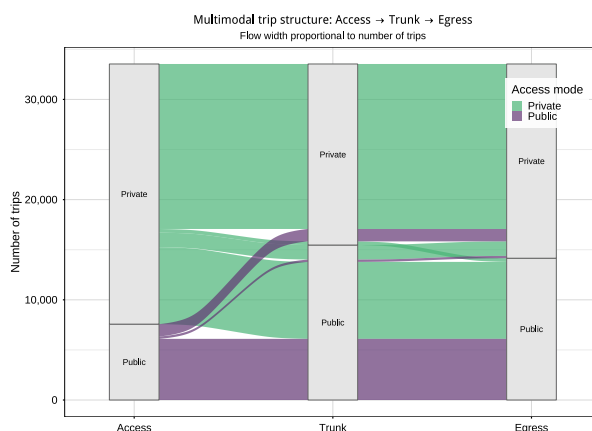
FIGURE 12. DISTRIBUTION OF COMMUTING TRIPS BY MODE COMPOSITION



Notes: Weekday work commutes from the 2017 EOD, excluding walk-only and non-transit trips. Trips are classified as private only (all segments served by combis), public only (all segments served by metro, BRT, bus, or train), or mixed (combining both).

Figure 13 traces the structure of multimodal trips by decomposing each journey into three stages—access, trunk, and egress—and tracking whether each stage is served by private or public transit. The dominant pattern is clear: private transit serves as the primary access and egress mode, feeding commuters into and out of public trunk services. The most common multimodal configuration involves private access, public trunk, and private egress, consistent with the first-and-last-mile role of combis documented in the main text.

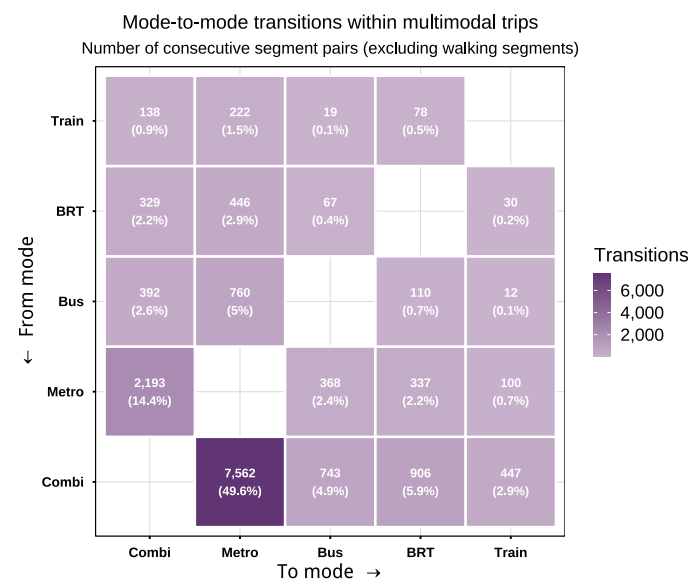
FIGURE 13. MULTIMODAL TRIP STRUCTURE: ACCESS, TRUNK, AND EGRESS



Notes: Alluvial diagram decomposing multimodal trips into access, trunk, and egress stages. Flow widths are proportional to the number of trips. Green denotes private (combi) and purple denotes public transit.

Mode-to-mode transitions. Figure 14 displays a transition matrix of consecutive mode pairs within multimodal trips, excluding walking segments. The single most frequent transition is from combi to metro, accounting for 49.6 percent of all observed transitions (7,562 pairs). The next largest combi transitions are combi-to-BRT (5.9 percent) and combi-to-bus (4.9 percent). Metro-to-combi is the dominant reverse transition (14.4 percent). These patterns confirm that private transit overwhelmingly serves as the feeder mode into public trunk infrastructure, with the combi-metro interface constituting the backbone of multimodal commuting in the ZMVM.

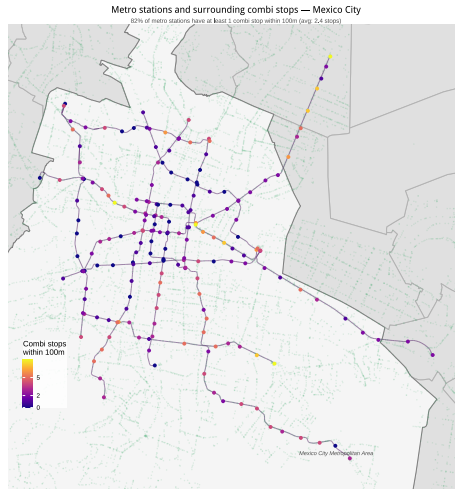
FIGURE 14. MODE-TO-MODE TRANSITIONS WITHIN MULTIMODAL TRIPS



Notes: Each cell reports the number and share of consecutive segment pairs transitioning from the row mode to the column mode, excluding walking segments. Darker shading indicates higher frequency.

Spatial co-location. Figure 15 maps every metro station in Mexico City, colored by the number of combi stops located within 100 meters. Eighty-two percent of metro stations have at least one combi stop in their immediate vicinity, with an average of 2.4 combi stops per station. This physical co-location reflects the organic emergence of private feeder services around public infrastructure and provides infrastructure-level confirmation of the trip-level complementarity documented above.

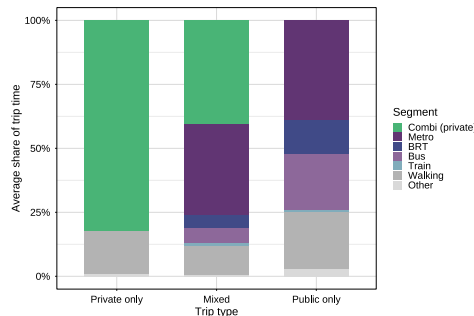
FIGURE 15. METRO STATIONS AND SURROUNDING COMBI STOPS



Notes: Each dot represents a metro station, colored by the number of combi stops within 100 meters. Eighty-two percent of stations have at least one combi stop nearby, with an average of 2.4 stops per station.

Time decomposition and trip characteristics. Figure 16 decomposes average trip time by mode segment across the three trip types. Private-only trips are dominated by combi time, while mixed trips allocate substantial shares to both combi and public modes (metro, BRT, bus, and train), confirming that private transit absorbs a large fraction of total travel time even on trips that use public infrastructure. Public-only trips are split primarily between metro and BRT segments.

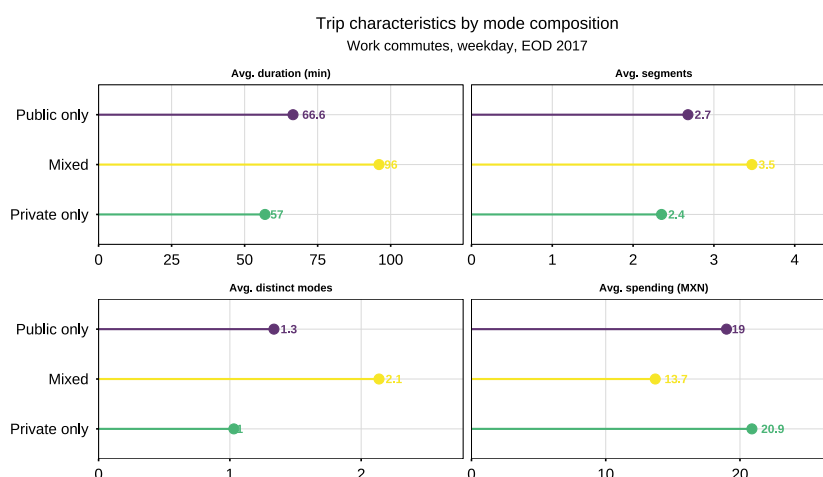
FIGURE 16. AVERAGE SHARE OF TRIP TIME BY MODE SEGMENT



Notes: Stacked bars show the average share of total trip time spent on each mode segment, separately for private-only, mixed, and public-only trips. Weekday work commutes, EOD 2017.

Figure 17 summarizes four trip attributes by mode composition. Mixed trips are the longest (96 minutes on average), involve the most segments (3.5) and distinct modes (2.1), and have intermediate monetary cost (13.7 MXN). Private-only trips are shorter (57 minutes) and less complex (2.4 segments, 1 mode) but the most expensive (20.9 MXN), reflecting higher per-kilometer costs of combi service. Public-only trips fall between the two in duration (66.6 minutes) and cost (19 MXN). These patterns are consistent with a system in which commuters combine modes to exploit the speed advantages of public trunk services while relying on private transit for spatial coverage—at the cost of longer and more complex journeys.

FIGURE 17. TRIP CHARACTERISTICS BY MODE COMPOSITION

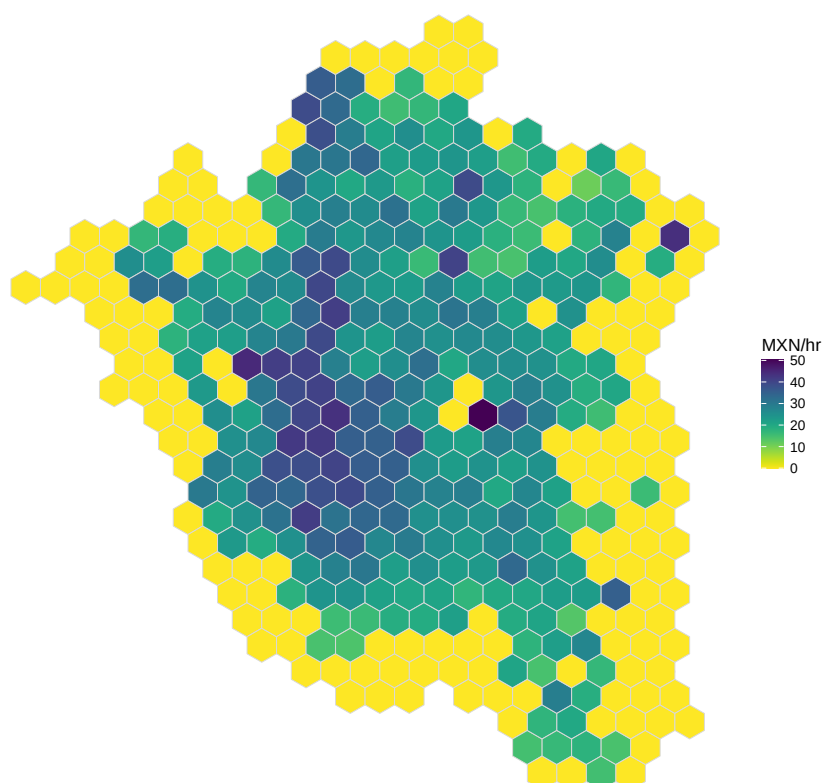


Notes: Average duration (minutes), number of segments, number of distinct modes, and monetary spending (MXN) for each trip type. Weekday work commutes, EOD 2017.

A.3 Productivity map

FIGURE 18. SPATIAL DISTRIBUTION OF PRODUCTIVITY IN MEXICO CITY

Productivity (hourly wage) -- 5 km hex grid
Source: Censos Económicos, area-weighted by AGEB



B SENSITIVITY ANALYSIS

TABLE 5—SENSITIVITY ANALYSIS: WELFARE ACROSS BUDGET INCREASES ($K_{\text{CALIB}} = 2000$)

ΔK (%)	K_B	W^{Base}	W^{B1}	W^{B2}	ΔW^{B1} (%)	ΔW^{B2} (%)	$\Delta W^{\text{B2-B1}}$ (%)
10	200	35.6793	35.8055	35.8988	0.3538	0.6154	0.2606
25	500	35.6793	35.9445	36.0380	0.7434	1.0056	0.2602
50	1000	35.6793	36.1553	36.2481	1.3342	1.5945	0.2568
75	1500	35.6793	36.3417	36.4319	1.8567	2.1094	0.2481

TABLE 6—SENSITIVITY ANALYSIS: WELFARE ACROSS θ_{PRI}

θ_{pri}	W^{Base}	W^{B1}	W^{B2}	ΔW^{B1} (%)	ΔW^{B2} (%)	$\Delta W^{\text{B2-B1}}$ (%)
1.05	35.7588	36.2295	36.3655	1.3164	1.6967	0.3754
2.04	35.6508	36.1295	36.2060	1.3427	1.5572	0.2117
3.04	35.6133	36.0959	36.1489	1.3551	1.5039	0.1468
4.03	35.5950	36.0794	36.1197	1.3610	1.4740	0.1116
5.03	35.5843	36.0694	36.1020	1.3633	1.4549	0.0904
6.02	35.5773	36.0636	36.0902	1.3670	1.4417	0.0737
7.02	35.5724	36.0593	36.0818	1.3688	1.4321	0.0625
8.01	35.5688	36.0561	36.0755	1.3702	1.4248	0.0539
9.01	35.5660	36.0537	36.0707	1.3714	1.4191	0.0471
10.00	35.5637	36.0518	36.0668	1.3724	1.4145	0.0415

TABLE 7—SENSITIVITY ANALYSIS: WELFARE ACROSS δ_{PRI} (WITH $K_B = 1.5 \times K_{\text{CALIB}}$)

δ_{pri}	K_{calib}	K_B	W^{Base}	W^{B1}	W^{B2}	ΔW^{B1} (%)	ΔW^{B2} (%)	$\Delta W^{\text{B2-B1}}$ (%)
0.0201	1147.2	1720.8	35.6793	35.9777	36.0908	0.8364	1.1533	0.3143
0.0515	1478.8	2218.3	35.6793	36.0490	36.1497	1.0364	1.3185	0.2792
0.0828	1810.5	2715.7	35.6793	36.1173	36.2124	1.2278	1.4943	0.2632
0.1142	2142.1	3213.2	35.6793	36.1835	36.2751	1.4132	1.6699	0.2531
0.1455	2473.8	3710.7	35.6793	36.2486	36.3370	1.5956	1.8435	0.2440
0.1768	2805.4	4208.1	35.6793	36.3084	36.3937	1.7634	2.0024	0.2349
0.2082	3137.1	4705.6	35.6793	36.3649	36.4484	1.9218	2.1557	0.2295
0.2395	3468.7	5203.1	35.6793	36.4204	36.5010	2.0773	2.3031	0.2211
0.2709	3800.4	5700.5	35.6793	36.4715	36.5499	2.2205	2.4402	0.2150
0.3022	4132.0	6198.0	35.6793	36.5210	36.5982	2.3591	2.5755	0.2114

C MODEL DERIVATIONS

C.1 First-order conditions of the planner's problem

This appendix derives the first-order conditions for the mode-specific formulation. Define

$$A_{jk}^{\text{pub}} \equiv \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}}, \quad A_{jk}^{\text{pri}} \equiv \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}}.$$

Then

$$T_{jk}^{\text{pub}} = A_{jk}^{\text{pub}} L_{jk}^{\text{pub}}, \quad T_{jk}^{\text{pri}} = A_{jk}^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}.$$

Let $\eta_{jk}^{\text{pub}} \geq 0$ be the multiplier on public capacity, written as $\kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}} - L_{jk}^{\text{pub}} \geq 0$. The Lagrangian is

$$\begin{aligned} \mathcal{L} = & u + \sum_{j \in \mathcal{J}} \omega_j L_j^R (U(c_j, h_j) - u) \\ & + \sum_{j \in \mathcal{J}} P_j^F \left(L_j^R + \sum_{i \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{ij}^m - L_j^L - \sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{jk}^m - \sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} T_{jk}^m \right) \\ & + P^R \left(\bar{L} - \sum_{j \in \mathcal{J}} L_j^R \right) \\ & + P^L \left(\bar{L} - \sum_{j \in \mathcal{J}} L_j^L - \sum_{(j,k) \in \mathcal{E}} \sum_{m \in \mathcal{M}} T_{jk}^m \right) \\ & + P^C \left(\sum_{j \in \mathcal{J}} A_j L_j^L - \sum_{j \in \mathcal{J}} c_j L_j^R \right) + \sum_{j \in \mathcal{J}} P_j^H (H_j - h_j L_j^R) \\ & + P^K \left(K - \sum_{(j,k) \in \mathcal{E}} \left[\delta_{jk}^{I, \text{pub}} I_{jk}^{\text{pub}} + \delta_{jk}^{I, \text{pri}} I_{jk}^{\text{pri}} \right] \right) \\ & + \sum_{(j,k) \in \mathcal{E}} \eta_{jk}^{\text{pub}} \left(\kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}} - L_{jk}^{\text{pub}} \right) \\ & + \sum_{j \in \mathcal{J}} \mu_j^c c_j + \sum_{j \in \mathcal{J}} \mu_j^h h_j + \sum_{j \in \mathcal{J}} \mu_j^R L_j^R + \sum_{j \in \mathcal{J}} \mu_j^L L_j^L \\ & + \sum_{(j,k) \in \mathcal{E}} \nu_{jk}^{\text{pub}} L_{jk}^{\text{pub}} + \sum_{(j,k) \in \mathcal{E}} \nu_{jk}^{\text{pri}} L_{jk}^{\text{pri}} \\ & + \sum_{(j,k) \in \mathcal{E}} \sum_{m \in \mathcal{M}} \mu_{jk}^{I, m} \left(I_{jk}^m - \bar{I}_{jk}^m \right) + \sum_{(j,k) \in \mathcal{E}} \sum_{m \in \mathcal{M}} \bar{\mu}_{jk}^{I, m} \left(\bar{I}_{jk}^m - I_{jk}^m \right). \end{aligned} \quad (21)$$

The first-order conditions with respect to consumption, housing, residents, and local labor are

$$[c_j] \quad \omega_j L_j^R U_c(c_j, h_j) + \mu_j^c = P^C L_j^R, \quad \forall j, \quad (22)$$

$$[h_j] \quad \omega_j L_j^R U_h(c_j, h_j) + \mu_j^h = P_j^H L_j^R, \quad \forall j, \quad (23)$$

$$[L_j^L] \quad -P_j^F - P^L + P^C A_j + \mu_j^L = 0, \quad \forall j, \quad (24)$$

$$[L_j^R] \quad \omega_j(U(c_j, h_j) - u) + P_j^F - P^R - P^C c_j - P_j^H h_j + \mu_j^R = 0, \quad \forall j. \quad (25)$$

Mode-specific flow conditions. For private flow, differentiating with respect to L_{jk}^{pri} gives

$$[L_{jk}^{\text{pri}}] \quad P_k^F - P_j^F - (P_j^F + P^L) \frac{\partial T_{jk}^{\text{pri}}}{\partial L_{jk}^{\text{pri}}} + \nu_{jk}^{\text{pri}} = 0. \quad (26)$$

Together with complementary slackness, this implies

$$P_k^F - P_j^F \leq (P_j^F + P^L)(1 + \beta) A_{jk}^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^\beta \left(I_{jk}^{\text{pri}} \right)^{-\gamma}, \quad = \text{ if } L_{jk}^{\text{pri}} > 0. \quad (27)$$

When the private mode is active, solving for L_{jk}^{pri} yields (11).

For public flow, differentiating with respect to L_{jk}^{pub} gives

$$[L_{jk}^{\text{pub}}] \quad P_k^F - P_j^F - (P_j^F + P^L) A_{jk}^{\text{pub}} - \eta_{jk}^{\text{pub}} + \nu_{jk}^{\text{pub}} = 0. \quad (28)$$

Thus

$$P_k^F - P_j^F \leq (P_j^F + P^L) A_{jk}^{\text{pub}} + \eta_{jk}^{\text{pub}}, \quad = \text{ if } L_{jk}^{\text{pub}} > 0. \quad (29)$$

Public infrastructure. Differentiating with respect to I_{jk}^{pub} gives

$$[I_{jk}^{\text{pub}}] \quad \eta_{jk}^{\text{pub}} \kappa_{jk}^{\text{pub}} - P^K \delta_{jk}^{I, \text{pub}} + \underline{\mu}_{jk}^{I, \text{pub}} - \bar{\mu}_{jk}^{I, \text{pub}} = 0. \quad (30)$$

An interior public-infrastructure choice therefore satisfies (12). Since $\eta_{jk}^{\text{pub}} > 0$ only when public capacity binds, public infrastructure is built because it relaxes the feasible-flow constraint for public riders, not because it lowers public riders' operating cost.

Private infrastructure. Differentiating with respect to I_{jk}^{pri} gives

$$[I_{jk}^{\text{pri}}] \quad - (P_j^F + P^L) \frac{\partial T_{jk}^{\text{pri}}}{\partial I_{jk}^{\text{pri}}} - P^K \delta_{jk}^{I, \text{pri}} + \underline{\mu}_{jk}^{I, \text{pri}} - \bar{\mu}_{jk}^{I, \text{pri}} = 0. \quad (31)$$

Because

$$\frac{\partial T_{jk}^{\text{pri}}}{\partial I_{jk}^{\text{pri}}} = -\gamma A_{jk}^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma-1}, \quad (32)$$

an interior solution satisfies (15), which yields (16).

C.2 Proof of Proposition 1

Fix an edge (j, k) and suppress edge subscripts whenever this creates no ambiguity. Define

$$V_j \equiv P_j^F + P^L, \quad A^{\text{pub}} \equiv \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}}, \quad A^{\text{pri}} \equiv \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}},$$

and let

$$B^{\text{pub}} \equiv P^K \delta_{jk}^{I, \text{pub}}, \quad B^{\text{pri}} \equiv P^K \delta_{jk}^{I, \text{pri}}$$

denote the shadow cost of building one unit of public and private infrastructure on link (j, k) . Let

$$\Delta P_{jk} \equiv P_k^F - P_j^F$$

be the destination potential on the edge. We prove the result for an interior solution with positive public and private flows and slack infrastructure bounds. Corner solutions are characterized by the corresponding complementary slackness inequalities.

The part of the Lagrangian that depends on the mode-specific flows and infrastructure on edge (j, k) is

$$\begin{aligned} \mathcal{L}_{jk} = & \Delta P_{jk} (L_{jk}^{\text{pub}} + L_{jk}^{\text{pri}}) \\ & - V_j \left[A^{\text{pub}} L_{jk}^{\text{pub}} + A^{\text{pri}} (L_{jk}^{\text{pri}})^{1+\beta} (I_{jk}^{\text{pri}})^{-\gamma} \right] \\ & - B^{\text{pub}} I_{jk}^{\text{pub}} - B^{\text{pri}} I_{jk}^{\text{pri}} \\ & - \eta_{jk}^{\text{pub}} (L_{jk}^{\text{pub}} - \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}), \end{aligned}$$

where $\eta_{jk}^{\text{pub}} \geq 0$ is the multiplier on the public-capacity constraint

$$L_{jk}^{\text{pub}} \leq \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}.$$

First consider public transit. The first-order condition with respect to public flow is

$$\frac{\partial \mathcal{L}_{jk}}{\partial L_{jk}^{\text{pub}}} = \Delta P_{jk} - V_j A^{\text{pub}} - \eta_{jk}^{\text{pub}} = 0,$$

whenever $L_{jk}^{\text{pub}} > 0$. Therefore,

$$\Delta P_{jk} = V_j A^{\text{pub}} + \eta_{jk}^{\text{pub}}. \tag{33}$$

The first-order condition with respect to public infrastructure is

$$\frac{\partial \mathcal{L}_{jk}}{\partial I_{jk}^{\text{pub}}} = \eta_{jk}^{\text{pub}} \kappa_{jk}^{\text{pub}} - B^{\text{pub}} = 0.$$

Thus,

$$\eta_{jk}^{\text{pub}} = \frac{B^{\text{pub}}}{\kappa_{jk}^{\text{pub}}} = \frac{P^K \delta_{jk}^{I, \text{pub}}}{\kappa_{jk}^{\text{pub}}}. \tag{34}$$

Since $B^{\text{pub}} > 0$ and $\kappa_{jk}^{\text{pub}} > 0$, equation (34) implies $\eta_{jk}^{\text{pub}} > 0$. By complementary slackness, public capacity must bind on an active public edge:

$$L_{jk}^{\text{pub}} = \kappa_{jk}^{\text{pub}} I_{jk}^{\text{pub}}. \quad (35)$$

Substituting (34) into (33), the generalized unit cost of public service is

$$c_{jk}^{\text{pub}} = V_j A^{\text{pub}} + \frac{B^{\text{pub}}}{\kappa_{jk}^{\text{pub}}}. \quad (36)$$

Returning to primitives,

$$c_{jk}^{\text{pub}} = (P_j^F + P^L) \delta_{jk}^{\tau, \text{pub}} \theta_{\text{pub}} + \frac{P^K \delta_{jk}^{I, \text{pub}}}{\kappa_{jk}^{\text{pub}}}.$$

The first term is the operating cost of serving one more public rider, valued in the planner's numeraire. The second term is the shadow cost of the public infrastructure required to create one more unit of public passenger capacity.

Now consider private transit. The first-order condition with respect to private infrastructure is

$$\frac{\partial \mathcal{L}_{jk}}{\partial I_{jk}^{\text{pri}}} = V_j \gamma A^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma-1} - B^{\text{pri}} = 0.$$

Solving for private infrastructure gives

$$I_{jk}^{\text{pri},*} = \left[\frac{\gamma V_j A^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta}}{B^{\text{pri}}} \right]^{1/(1+\gamma)}. \quad (37)$$

In primitives, this is

$$I_{jk}^{\text{pri},*} = \left[\frac{\gamma (P_j^F + P^L) \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{1+\beta}}{P^K \delta_{jk}^{I, \text{pri}}} \right]^{1/(1+\gamma)}.$$

The first-order condition with respect to private flow is

$$\frac{\partial \mathcal{L}_{jk}}{\partial L_{jk}^{\text{pri}}} = \Delta P_{jk} - V_j (1 + \beta) A^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma} = 0,$$

whenever $L_{jk}^{\text{pri}} > 0$. Hence,

$$\Delta P_{jk} = V_j (1 + \beta) A^{\text{pri}} \left(L_{jk}^{\text{pri}} \right)^{\beta} \left(I_{jk}^{\text{pri}} \right)^{-\gamma}. \quad (38)$$

Substituting (37) into (38) yields the optimized generalized marginal cost of private service:

$$c_{jk}^{\text{pri}}(L_{jk}^{\text{pri}}) = (1 + \beta)\gamma^{-\gamma/(1+\gamma)} (V_j A^{\text{pri}})^{1/(1+\gamma)} (B^{\text{pri}})^{\gamma/(1+\gamma)} (L_{jk}^{\text{pri}})^{(\beta-\gamma)/(1+\gamma)}. \quad (39)$$

Returning to primitives,

$$c_{jk}^{\text{pri}}(L_{jk}^{\text{pri}}) = (1 + \beta)\gamma^{-\gamma/(1+\gamma)} \left[(P_j^F + P^L) \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \right]^{1/(1+\gamma)} \left[P^K \delta_{jk}^{I, \text{pri}} \right]^{\gamma/(1+\gamma)} (L_{jk}^{\text{pri}})^{(\beta-\gamma)/(1+\gamma)}.$$

Finally, if both public and private modes are used on the same edge, then both first-order conditions must hold with equality and both modes deliver the same marginal benefit ΔP_{jk} . Combining (33) and (38), after substituting the infrastructure first-order conditions, gives

$$c_{jk}^{\text{pub}} = c_{jk}^{\text{pri}}(L_{jk}^{\text{pri}}). \quad (40)$$

This proves the proposition.

The expression also shows how the technology comparison depends on the curvature of the private technology. If $\beta > \gamma$, then

$$\frac{\beta - \gamma}{1 + \gamma} > 0,$$

so the optimized private marginal cost is increasing in private flow. Public transit then has a threshold interpretation: private transit is attractive for low-flow links, but high-flow corridors eventually justify public capacity. If $\beta = \gamma$, then

$$c_{jk}^{\text{pri}} = (1 + \beta)\beta^{-\beta/(1+\beta)} \left[(P_j^F + P^L) \delta_{jk}^{\tau, \text{pri}} \theta_{\text{pri}} \right]^{1/(1+\beta)} \left[P^K \delta_{jk}^{I, \text{pri}} \right]^{\beta/(1+\beta)},$$

which is independent of private flow. In that boundary case, both public and private technologies have constant effective unit-cost indices, and equality of these indices can generate multiple optimal public-private allocations with the same welfare value.

C.3 Proof of Proposition 2

We prove Proposition 2. Consider first the private transport block. Suppressing edge subscripts, write

$$T(L, I) = AL^{1+\beta}I^{-\gamma}, \quad A > 0.$$

The Hessian of T with respect to (L, I) is

$$\nabla^2 T(L, I) = A \begin{pmatrix} \beta(1 + \beta)L^{\beta-1}I^{-\gamma} & -\gamma(1 + \beta)L^{\beta}I^{-\gamma-1} \\ -\gamma(1 + \beta)L^{\beta}I^{-\gamma-1} & \gamma(1 + \gamma)L^{1+\beta}I^{-\gamma-2} \end{pmatrix}.$$

The leading principal minor is non-negative when $\beta \geq 0$. The determinant is

$$\det \nabla^2 T(L, I) = A^2(1 + \beta)\gamma(\beta - \gamma)L^{2\beta}I^{-2\gamma-2}.$$

Therefore $\nabla^2 T(L, I)$ is positive semidefinite whenever $\beta \geq \gamma \geq 0$. Hence the private transport block is convex.

If $\beta = \gamma > 0$, then

$$T(L, I) = AL^{1+\beta}I^{-\beta}$$

is homogeneous of degree one:

$$T(tL, tI) = tT(L, I).$$

Thus the function has zero curvature along rays $(L, I) \mapsto (tL, tI)$ and cannot be strictly or strongly convex. This proves that the private block is weakly convex at $\beta = \gamma$.

The public block is linear in public flow,

$$T^{\text{pub}}(L^{\text{pub}}) = A^{\text{pub}}L^{\text{pub}},$$

and the public-capacity constraint

$$L^{\text{pub}} \leq \kappa^{\text{pub}}I^{\text{pub}}$$

is linear. The network-building constraint, infrastructure bounds, non-negativity constraints, and population constraint are also linear.

It remains to note that the balanced-flow constraint is convex once the transport block is convex. For each node j , the constraint can be written as

$$L_j^L + \sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{jk}^m + \sum_{k \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} T_{jk}^m - L_j^R - \sum_{i \in \mathcal{N}(j)} \sum_{m \in \mathcal{M}} L_{ij}^m \leq 0.$$

This is a sum of linear terms and convex transport resource costs. Therefore it is convex. Since all remaining constraints are linear or convex, and the objective is linear in u , the full planner's problem is convex under the standard aggregate-variable representation. With mobile labor and homothetic preferences, the same argument delivers the usual quasiconvex formulation used by [Fajgelbaum and Schaal \(2020\)](#).